

DAA LAB

CSA0601

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TOPIC 1 – INTRODUCTION

1. Bubble Sort

Aim:

To sort the given array in ascending order using Bubble Sort.

Algorithm:

1. Read the array elements.
2. Compare adjacent elements.
3. Swap if they are in the wrong order.
4. Repeat until the array is sorted.
5. Display the sorted array.

Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a[100], n, i, j, temp;
```

```
    printf("Enter the number of elements: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter the elements:\n");
```

```
    for(i = 0; i < n; i++)
```

```
scanf("%d", &a[i]);

for(i = 0; i < n - 1; i++)
{
    for(j = 0; j < n - i - 1; j++)
    {
        if(a[j] > a[j + 1])
        {
            temp = a[j];
            a[j] = a[j + 1];
            a[j + 1] = temp;
        }
    }
}
```

```
printf("Sorted array: ");

for(i = 0; i < n; i++)

    printf("%d ", a[i]);
```

```
return 0;

}
```

Output

```
#include <stdio.h>

int main()
{
    int a[100], n, i, j, temp;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter the elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    // Bubble Sort
    for(i = 0; i < n - 1; i++)
    {
        for(j = 0; j < n - i - 1; j++)
        {
            if(a[j] > a[j + 1])
            {
                temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;
            }
        }
    }

    printf("Sorted array: ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);
    printf("\n");
}
```

Enter the number of elements: 5
Enter the elements:
23 45 67 54 43
Sorted array: 23 43 45 54 67
=== Code Execution Successful ===

Result:

The array is sorted successfully using Bubble Sort.

2. Find Maximum Element

Aim:

To find the maximum element in the given array.

Algorithm:

1. Read the array elements.
2. Assume the first element is maximum.
3. Compare it with the remaining elements.
4. Update the maximum if a larger element is found.
5. Display the maximum element.

Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a[100],n,i,max;
```

```
printf("Enter number of elements: ");
```

```
scanf("%d",&n);
```

```
printf("Enter elements:\n");
```

```
for(i=0;i<n;i++)
```

```
    scanf("%d",&a[i]);
```

```
max=a[0];
```

```
for(i=1;i<n;i++)
```

```
{
```

```
    if(a[i]>max)
```

```
        max=a[i];
```

```
}
```

```
printf("Maximum element = %d",max);
```

```
return 0;
```

```
}
```

Output

```
main.c | Run | Output
4+ {
5   int a[100], n, i, max;
6
7   printf("Enter the number of elements: ");
8   scanf("%d", &n);
9
10  printf("Enter the elements:\n");
11  for(i = 0; i < n; i++)
12      scanf("%d", &a[i]);
13
14  max = a[0];
15
16  for(i = 1; i < n; i++)
17  {
18      if(a[i] > max)
19          max = a[i];
20  }
21
22  printf("Maximum element = %d", max);
23
24  return 0;
25 }
```

Enter the number of elements: 5
5
Enter the elements:
45 56 74 23
Maximum element = 74
=== Code Execution Successful ===

Result:

The maximum element is found successfully.

3. Sort and Find Maximum

Aim:

To sort the array and find the maximum element.

Algorithm:

1. Read the array.
2. Sort it using Bubble Sort.
3. The last element is the maximum.
4. Display the sorted array and maximum element.

Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a[100],n,i,j,temp;
```

```
    printf("Enter number of elements: ");
```

```
    scanf("%d",&n);
```

```
printf("Enter elements:\n");
```

```
for(i=0;i<n;i++)
```

```
    scanf("%d",&a[i]);
```

```
for(i=0;i<n-1;i++)
```

```
{
```

```
    for(j=0;j<n-i-1;j++)
```

```
    {
```

```
        if(a[j]>a[j+1])
```

```
        {
```

```
            temp=a[j];
```

```
            a[j]=a[j+1];
```

```
            a[j+1]=temp;
```

```
        }
```

```
    }
```

```
}
```

```
printf("Sorted array:\n");
```

```
for(i=0;i<n;i++)
```

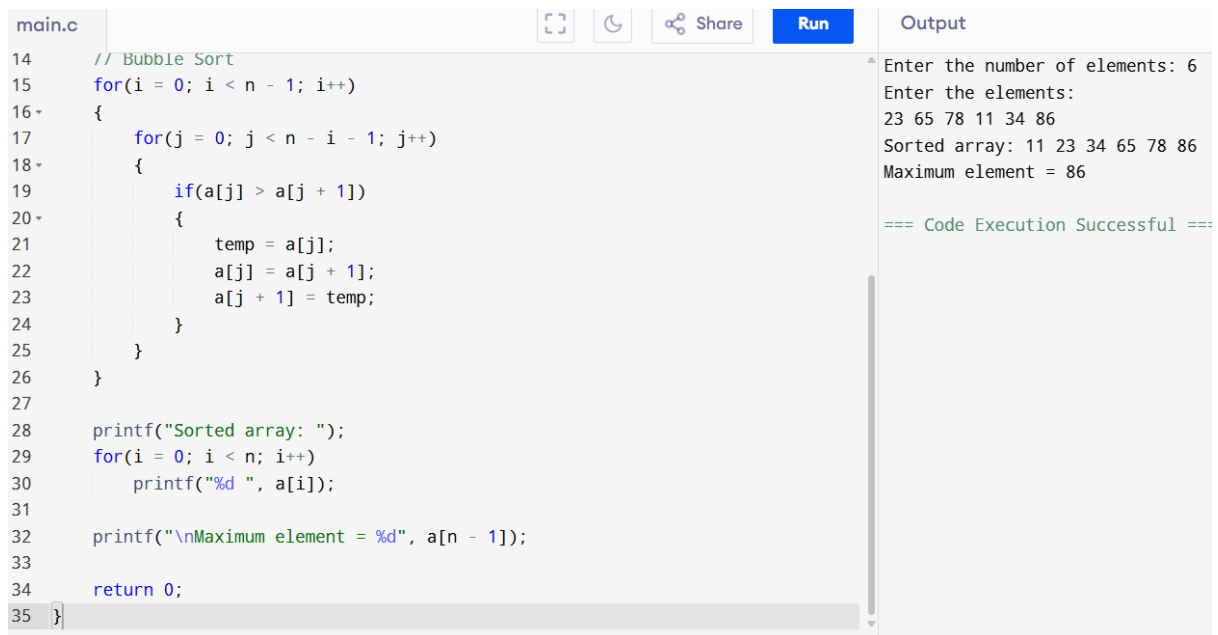
```
    printf("%d ",a[i]);
```

```
printf("\nMaximum element = %d",a[n-1]);
```

```
return 0;
```

```
}
```

Output



The screenshot shows a C program in a code editor with a 'Run' button. The code implements a Bubble Sort algorithm. The output window shows the following text:

```
Enter the number of elements: 6
Enter the elements:
23 65 78 11 34 86
Sorted array: 11 23 34 65 78 86
Maximum element = 86

=== Code Execution Successful ===
```

Result:

The array is sorted and the maximum element is displayed successfully.

4. Binary Search

Aim:

To search an element using Binary Search.

Algorithm:

1. Read the sorted array.
2. Find the middle element.
3. Compare it with the key.
4. Search the left or right half accordingly.
5. Display whether the element is found.

Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int a[100],n,i,key,low,high,mid;
```

```
printf("Enter number of elements: ");
```

```
scanf("%d",&n);
```

```
printf("Enter sorted elements:\n");
```

```
for(i=0;i<n;i++)
```

```
    scanf("%d",&a[i]);
```

```
printf("Enter element to search: ");
```

```
scanf("%d",&key);
```

```
low=0;
```

```
high=n-1;
```

```
while(low<=high)
```

```
{
```

```
    mid=(low+high)/2;
```

```
    if(a[mid]==key)
```

```
    {
```

```
        printf("Element found at position %d",mid+1);
```

```
        return 0;
```

```
    }
```

```
    else if(a[mid]<key)
```



```

        low=mid+1;

    else

        high=mid-1;

    }

    printf("Element not found");

    return 0;

}

```

Output

```

low = 0,
high = n - 1;

while(low <= high)
{
    mid = (low + high) / 2;

    if(a[mid] == key)
    {
        printf("Element found at position %d", mid + 1);
        return 0;
    }
    else if(a[mid] < key)
        low = mid + 1;
    else
        high = mid - 1;
}

printf("Element not found");

return 0;

```

```

Enter the number of elements: 7
Enter the sorted elements:
23 11 56 43 78 54 23
Enter the element to search: 56
Element not found

=== Code Execution Successful ===

```

Result:

The required element is searched successfully using Binary Search.

5. Efficient Sorting (Quick Sort)

Aim:

To sort the array using Quick Sort.

Algorithm:

1. Select a pivot element.

2. Partition the array.
3. Recursively sort both partitions.
4. Display the sorted array.

Program

```
#include <stdio.h>
```

```
void quicksort(int a[], int low, int high);
```

```
int partition(int a[], int low, int high);
```

```
int main()
```

```
{
```

```
    int a[100], n, i;
```

```
    printf("Enter number of elements: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter elements:\n");
```

```
    for(i = 0; i < n; i++)
```

```
        scanf("%d", &a[i]);
```

```
    quicksort(a, 0, n - 1);
```

```
    printf("Sorted array:\n");
```

```
    for(i = 0; i < n; i++)
```

```
        printf("%d ", a[i]);
```

```
    return 0;
}

void quicksort(int a[], int low, int high)
{
    int p;
    if(low < high)
    {
        p = partition(a, low, high);
        quicksort(a, low, p - 1);
        quicksort(a, p + 1, high);
    }
}
```

```
int partition(int a[], int low, int high)
{
    int pivot = a[high];
    int i = low - 1, j, temp;

    for(j = low; j < high; j++)
    {
        if(a[j] < pivot)
        {
            i++;
            temp = a[i];
```

```

        a[i] = a[j];

        a[j] = temp;

    }

}

temp = a[i + 1];

a[i + 1] = a[high];

a[high] = temp;

return i + 1;

}

```

Output

```

int partition(int a[], int low, int high)
{
    int pivot = a[high];
    int i = low - 1, j, temp;

    for(j = low; j < high; j++)
    {
        if(a[j] < pivot)
        {
            i++;
            temp = a[i];
            a[i] = a[j];
            a[j] = temp;
        }
    }

    temp = a[i + 1];
    a[i + 1] = a[high];
    a[high] = temp;

    return i + 1;
}

```

Enter the number of elements: 4
Enter the elements:
23 56 54 73
Sorted array: 23 54 56 73
=== Code Execution Successful ===

Result:

The array is sorted successfully using Quick Sort.

6. String Comparison Using Stack

Aim:

To compare two strings using stacks.

Algorithm:

1. Read two strings.
2. Push each string into separate stacks.
3. Compare the characters.
4. Display whether the strings are equal or not.

Program

```
#include <stdio.h>
```

```
#include <string.h>
```

```
char stack1[100], stack2[100];
```

```
int top1=-1, top2=-1;
```

```
void push1(char ch)
```

```
{
```

```
    stack1[++top1]=ch;
```

```
}
```

```
void push2(char ch)
```

```
{
```

```
    stack2[++top2]=ch;
```

```
}
```

```
int main()
```

```
{
```

```
    char str1[100], str2[100];
```

```
    int i;
```

```
    printf("Enter first string: ");
```

```
scanf("%s",str1);
```

```
printf("Enter second string: ");
```

```
scanf("%s",str2);
```

```
for(i=0; str1[i]!='\0'; i++)
```

```
    push1(str1[i]);
```

```
for(i=0; str2[i]!='\0'; i++)
```

```
    push2(str2[i]);
```

```
if(top1!=top2)
```

```
{
```

```
    printf("Strings are not equal");
```

```
    return 0;
```

```
}
```

```
while(top1>=0)
```

```
{
```

```
    if(stack1[top1]!=stack2[top2])
```

```
    {
```

```
        printf("Strings are not equal");
```

```
        return 0;
```

```
    }
```

```
    top1--;
```

```

        top2--;
    }

    printf("Strings are equal");

    return 0;
}

```

Output

The screenshot shows a code editor with C code on the left and its output on the right. The code defines two stacks, pushes 'hello' and 'world' onto them, and then compares them. Since the strings are not equal, it prints 'Strings are not equal'. The output window on the right shows the user input 'hello' and 'world', followed by the program's output 'Strings are not equal' and a success message.

```

if(top1 != top2)
{
    printf("Strings are not equal");
    return 0;
}

while(top1 >= 0)
{
    if(stack1[top1] != stack2[top2])
    {
        printf("Strings are not equal");
        return 0;
    }
    top1--;
    top2--;
}

printf("Strings are equal");

return 0;

```

Enter first string: hello
Enter second string: world
Strings are not equal
=== Code Execution Successful ===

Result:

The strings are compared successfully using stacks.

TOPIC 2 – BRUTE FORCE

1. List Operations

Aim:

To perform insertion, deletion, and display operations on a list.

Algorithm:

1. Read the list elements.
2. Select the required operation.
3. Perform insertion or deletion.

4. Display the updated list.

Program

```
#include <stdio.h>

int main()
{
    int a[100], n, i, pos, ele, choice;

    printf("Enter number of elements: ");
    scanf("%d",&n);

    printf("Enter elements:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("1.Insert\n2.Delete\n3.Display\n");
    printf("Enter choice: ");
    scanf("%d",&choice);

    switch(choice)
    {
        case 1:
            printf("Enter position and element: ");
            scanf("%d%d",&pos,&ele);
```



```
for(i=n;i>=pos;i--)
```

```
    a[i]=a[i-1];
```

```
    a[pos-1]=ele;
```

```
    n++;
```

```
    break;
```

```
case 2:
```

```
    printf("Enter position: ");
```

```
    scanf("%d",&pos);
```

```
    for(i=pos-1;i<n-1;i++)
```

```
        a[i]=a[i+1];
```

```
    n--;
```

```
    break;
```

```
}
```

```
printf("List elements:\n");
```

```
for(i=0;i<n;i++)
```

```
    printf("%d ",a[i]);
```

```
return 0;
```

```
}
```

Output

```
#include <stdio.h>

int main()
{
    int a[100], n, i, pos, ele, choice;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("\n1.Insert\n2.Delete\n3.Display\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            printf("Enter position and element: ");
            scanf("%d%d", &pos, &ele);
```

Enter number of elements: 5
Enter elements:
43 23 10 26 33

1.Insert
2.Delete
3.Display
Enter your choice: 1
Enter position and element: 2 23
List elements are:
43 23 23 10 26 33

=== Code Execution Successful ===

Result:

The list operations are performed successfully.

2. Selection Sort

Aim:

To sort the array using Selection Sort.

Algorithm:

1. Find the minimum element.
2. Swap it with the first unsorted element.
3. Repeat until sorted.
4. Display the sorted array.

Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a[100], n, i, j, min, temp;
```

```
    printf("Enter number of elements: ");
```

```
scanf("%d",&n);
```

```
printf("Enter elements:\n");
```

```
for(i=0;i<n;i++)
```

```
    scanf("%d",&a[i]);
```

```
for(i=0;i<n-1;i++)
```

```
{
```

```
    min=i;
```

```
    for(j=i+1;j<n;j++)
```

```
    {
```

```
        if(a[j]<a[min])
```

```
            min=j;
```

```
    }
```

```
    temp=a[i];
```

```
    a[i]=a[min];
```

```
    a[min]=temp;
```

```
}
```

```
printf("Sorted array:\n");
```

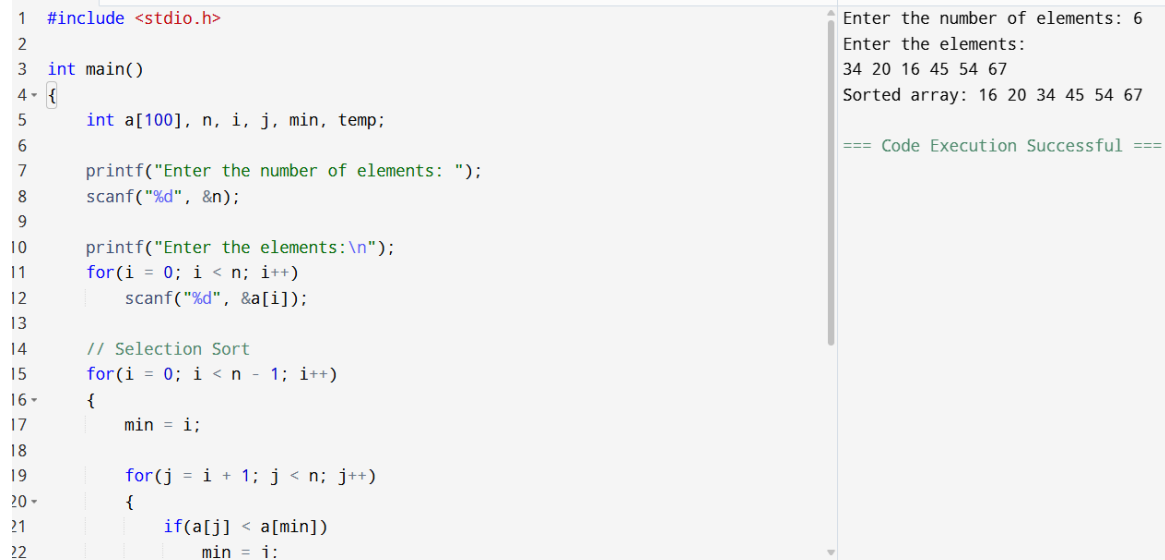
```
for(i=0;i<n;i++)
```

```
    printf("%d ",a[i]);
```

```
    return 0;

}
```

Output



```
1  #include <stdio.h>
2
3  int main()
4  {
5      int a[100], n, i, j, min, temp;
6
7      printf("Enter the number of elements: ");
8      scanf("%d", &n);
9
10     printf("Enter the elements:\n");
11     for(i = 0; i < n; i++)
12     {
13         scanf("%d", &a[i]);
14     }
15     // Selection Sort
16     for(i = 0; i < n - 1; i++)
17     {
18         min = i;
19
20         for(j = i + 1; j < n; j++)
21         {
22             if(a[j] < a[min])
23                 min = j;
24         }
25         temp = a[i];
26         a[i] = a[min];
27         a[min] = temp;
28     }
29     printf("Sorted array: ");
30     for(i = 0; i < n; i++)
31     {
32         printf("%d ", a[i]);
33     }
34     printf("\n");
35 }
```

Enter the number of elements: 6
Enter the elements:
34 20 16 45 54 67
Sorted array: 16 20 34 45 54 67
=== Code Execution Successful ===

Result:

The array is sorted successfully using Selection Sort.

3. Insertion Sort

Aim:

To sort the array using Insertion Sort.

Algorithm:

1. Assume the first element is sorted.
2. Insert each remaining element into its correct position.
3. Repeat for all elements.
4. Display the sorted array.

Program

```
#include <stdio.h>
```

```
int main()
```

```
{  
  
    int a[100], n, i, j, key;  
  
    printf("Enter number of elements: ");  
    scanf("%d",&n);  
  
    printf("Enter elements:\n");  
    for(i=0;i<n;i++)  
        scanf("%d",&a[i]);  
  
    for(i=1;i<n;i++)  
    {  
        key=a[i];  
        j=i-1;  
  
        while(j>=0 && a[j]>key)  
        {  
            a[j+1]=a[j];  
            j--;  
        }  
  
        a[j+1]=key;  
    }  
  
    printf("Sorted array:\n");
```

```

        for(i=0;i<n;i++)

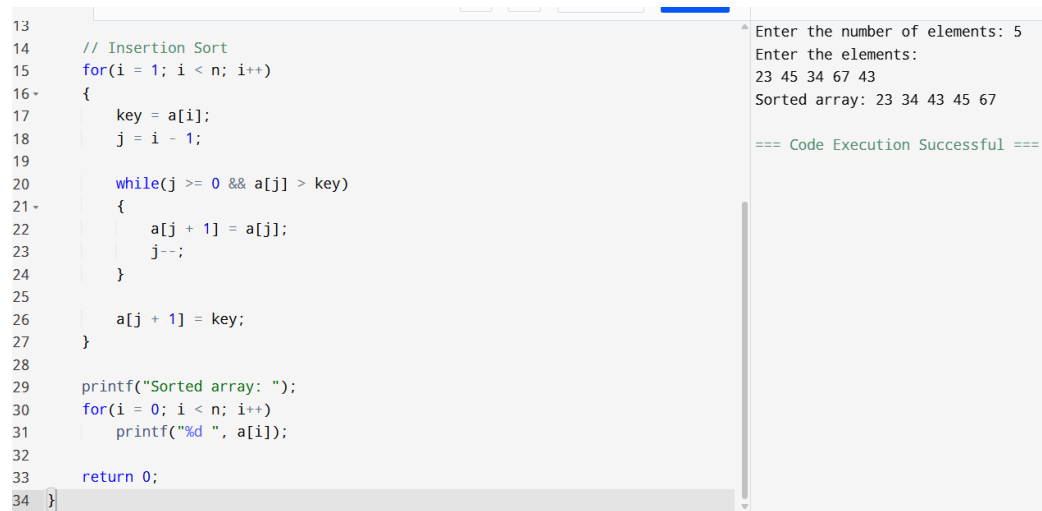
            printf("%d ",a[i]);


    return 0;

}

```

Output



The screenshot shows a code editor with C code for Insertion Sort. The code is as follows:

```

13
14 // Insertion Sort
15 for(i = 1; i < n; i++)
16 {
17     key = a[i];
18     j = i - 1;
19
20     while(j >= 0 && a[j] > key)
21     {
22         a[j + 1] = a[j];
23         j--;
24     }
25
26     a[j + 1] = key;
27 }
28
29 printf("Sorted array: ");
30 for(i = 0; i < n; i++)
31     printf("%d ", a[i]);
32
33 return 0;
34 }

```

The output on the right side of the editor shows the following:

```

Enter the number of elements: 5
Enter the elements:
23 45 34 67 43
Sorted array: 23 34 43 45 67

=== Code Execution Successful ===

```

Result:

The array is sorted successfully using Insertion Sort.

4. Optimized Bubble Sort

Aim:

To sort the array using Optimized Bubble Sort.

Algorithm:

1. Compare adjacent elements.
2. Swap if necessary.
3. Use a flag to detect swaps.
4. Stop if no swaps occur.
5. Display the sorted array.

Program

```
#include <stdio.h>
```

```
int main()

{

    int a[100], n, i, j, temp, swapped;


    printf("Enter number of elements: ");

    scanf("%d",&n);


    printf("Enter elements:\n");

    for(i=0;i<n;i++)

        scanf("%d",&a[i]);


    for(i=0;i<n-1;i++)

    {

        swapped=0;


        for(j=0;j<n-i-1;j++)

        {

            if(a[j]>a[j+1])

            {

                temp=a[j];

                a[j]=a[j+1];

                a[j+1]=temp;

                swapped=1;

            }

        }

    }

}
```

```

    }

    if(swapped==0)

        break;

}

printf("Sorted array:\n");

for(i=0;i<n;i++)

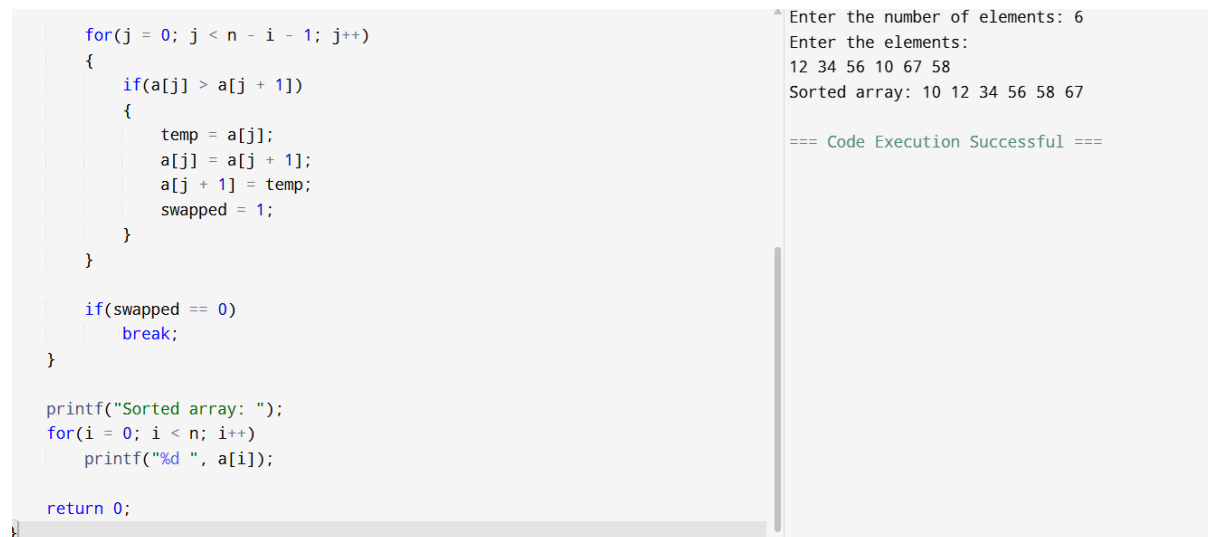
    printf("%d ",a[i]);

return 0;

}

```

Output



The screenshot shows a code editor with C code for an optimized bubble sort. The code includes a nested loop for swapping adjacent elements, a break statement when no swaps occur, and a final printf statement to display the sorted array. To the right of the code, the program's output is shown, including user input for the number of elements and the array values, followed by the sorted array and a success message.

```

for(j = 0; j < n - i - 1; j++)
{
    if(a[j] > a[j + 1])
    {
        temp = a[j];
        a[j] = a[j + 1];
        a[j + 1] = temp;
        swapped = 1;
    }
}

if(swapped == 0)
    break;

printf("Sorted array: ");
for(i = 0; i < n; i++)
    printf("%d ", a[i]);

return 0;

```

Enter the number of elements: 6
Enter the elements:
12 34 56 10 67 58
Sorted array: 10 12 34 56 58 67
=== Code Execution Successful ===

Result:

The array is sorted successfully using Optimized Bubble Sort.

5. Closest Pair of Points

Aim:

To find the closest pair of points using the brute-force method.

Algorithm:

1. Read all points.
2. Calculate the distance between every pair.
3. Find the minimum distance.
4. Display the closest distance.

Program

```
#include <stdio.h>
```

```
#include <math.h>
```

```
struct Point
```

```
{
```

```
    int x, y;
```

```
};
```

```
int main()
```

```
{
```

```
    struct Point p[100];
```

```
    int n, i, j;
```

```
    float dist, min;
```

```
    printf("Enter number of points: ");
```

```
    scanf("%d",&n);
```

```
    printf("Enter coordinates:\n");
```

```

for(i=0;i<n;i++)

    scanf("%d%d",&p[i].x,&p[i].y);


min=sqrt(pow(p[0].x-p[1].x,2)+pow(p[0].y-p[1].y,2));


for(i=0;i<n;i++)
{
    for(j=i+1;j<n;j++)
    {
        dist=sqrt(pow(p[i].x-p[j].x,2)+pow(p[i].y-p[j].y,2));

        if(dist<min)
            min=dist;
    }
}

printf("Minimum Distance = %.2f",min);


return 0;
}

```

Output

```

20 {
21     scanf("%d %d", &p[i].x, &p[i].y);
22 }
23
24 minDist = sqrt(pow(p[0].x - p[1].x, 2) + pow(p[0].y - p[1].y, 2));
25
26 for(i = 0; i < n; i++)
27 {
28     for(j = i + 1; j < n; j++)
29     {
30         dist = sqrt(pow(p[i].x - p[j].x, 2) + pow(p[i].y - p[j].y, 2));
31
32         if(dist < minDist)
33             minDist = dist;
34     }
35 }
36
37 printf("Minimum Distance = %.2f", minDist);
38
39 return 0;
40 }

```

Enter the number of points: 4
 Enter the coordinates (x y):
 2 3
 12 30
 40 50
 5 1
 Minimum Distance = 3.61
 === Code Execution Successful ===

Result:

The closest pair of points is found successfully.

6. Convex Hull (Brute Force)

Aim:

To find the Convex Hull using the brute-force method.

Algorithm:

1. Read all points.
2. Consider every pair of points.
3. Check whether all other points lie on the same side.
4. Display the Convex Hull edges.

Program

```
#include <stdio.h>
```

```
struct Point
```

```

{
    int x, y;
};

```

```
int main()
```

```

{

    struct Point p[100];

    int n, i, j, k;


    printf("Enter number of points: ");

    scanf("%d",&n);


    printf("Enter coordinates:\n");


    for(i=0;i<n;i++)

        scanf("%d%d",&p[i].x,&p[i].y);


    printf("Convex Hull Edges:\n");


    for(i=0;i<n;i++)

    {

        for(j=i+1;j<n;j++)

        {

            int left=0,right=0;


            for(k=0;k<n;k++)

            {

                int value=(p[j].x-p[i].x)*(p[k].y-p[i].y)-

                    (p[j].y-p[i].y)*(p[k].x-p[i].x);

```

```

        if(value>0)

            left++;

        else if(value<0)

            right++;

    }

    if(left==0 || right==0)

    {

        printf("(%d,%d) -> (%d,%d)\n",

            p[i].x,p[i].y,p[j].x,p[j].y);

    }

}

}

return 0;

}

```

Output

<pre> for(k = 0; k < n; k++) { int value = (p[j].x - p[i].x) * (p[k].y - p[i].y) - (p[j].y - p[i].y) * (p[k].x - p[i].x); if(value > 0) left++; else if(value < 0) right++; } if(left == 0 right == 0) { printf("(%d,%d) -> (%d,%d)\n", p[i].x, p[i].y, p[j].x, p[j].y); } </pre>	<pre> ^ Enter the number of points: 0 0 Enter the coordinates (x y): Convex Hull Edges: === Code Execution Successful === </pre>
--	---

Result:

The Convex Hull is obtained successfully using the brute-force method.

7. Convex Hull Program (Jarvis March)

Aim:

To find the Convex Hull using Jarvis March algorithm.

Algorithm:

1. Find the leftmost point.
2. Select the next counter-clockwise point.
3. Repeat until the starting point is reached.
4. Display the Convex Hull.

Program

```
#include <stdio.h>
```

```
struct Point
```

```
{
```

```
    int x, y;
```

```
};
```

```
int orientation(struct Point p, struct Point q, struct Point r)
```

```
{
```

```
    int val = (q.y - p.y) * (r.x - q.x) -
```

```
              (q.x - p.x) * (r.y - q.y);
```

```
    if(val == 0)
```

```
        return 0;
```

```
    return (val > 0) ? 1 : 2;
```

```
}
```

```

void convexHull(struct Point points[], int n)
{
    if(n < 3)
        return;

    int hull[100];

    int left = 0, p, q, i, count = 0;

    for(i = 1; i < n; i++)
        if(points[i].x < points[left].x)
            left = i;

    p = left;

    do
    {
        hull[count++] = p;

        q = (p + 1) % n;

        for(i = 0; i < n; i++)
            if(orientation(points[p], points[i], points[q]) == 2)
                q = i;

        p = q;
    } while(p != left);
}

```

```

    } while(p != left);

    printf("Convex Hull Points:\n");

    for(i = 0; i < count; i++)
        printf("(%d,%d)\n", points[hull[i]].x, points[hull[i]].y);
}

int main()
{
    struct Point points[100];
    int n, i;

    printf("Enter number of points: ");
    scanf("%d", &n);

    printf("Enter coordinates:\n");
    for(i = 0; i < n; i++)
        scanf("%d%d", &points[i].x, &points[i].y);

    convexHull(points, n);

    return 0;
}

```

Output


```
3     printf("Points on Convex Hull:\n");
4     for (int i = 0; i < count; i++)
5         printf("(%d, %d)\n", points[hull[i]].x, points[hull[i]].y);
6 }
7
8 int main()
9 {
10     int n;
11
12     printf("Enter number of points: ");
13     scanf("%d", &n);
14
15     struct Point points[100];
16
17     printf("Enter the coordinates (x y):\n");
18     for (int i = 0; i < n; i++)
19         scanf("%d %d", &points[i].x, &points[i].y);
20
21     convexHull(points, n);
22
23     return 0;
24 }
```

Enter number of points: 0 0
Enter the coordinates (x y):

=== Code Execution Successful ===

Result:

The Convex Hull is obtained successfully using Jarvis March.

8. Travelling Salesman Problem

Aim:

To find the minimum cost path for visiting all cities.

Algorithm:

1. Read the cost matrix.
2. Start from the first city.
3. Visit the nearest unvisited city.
4. Return to the starting city.
5. Display the minimum cost.

Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int cost[10][10], visited[10];
```

```
    int n, i, j, city = 0, next;
```

```
    int min, mincost = 0;
```

```
printf("Enter number of cities: ");
```

```
scanf("%d", &n);
```

```
printf("Enter cost matrix:\n");
```

```
for(i = 0; i < n; i++)
```

```
{
```

```
    visited[i] = 0;
```

```
    for(j = 0; j < n; j++)
```

```
        scanf("%d", &cost[i][j]);
```

```
}
```

```
visited[0] = 1;
```

```
printf("Path: 1");
```

```
for(i = 0; i < n - 1; i++)
```

```
{
```

```
    min = 999;
```

```
    for(j = 0; j < n; j++)
```

```
    {
```

```
        if(!visited[j] && cost[city][j] < min)
```

```

        {
            min = cost[city][j];
            next = j;
        }
    }

    visited[next] = 1;
    mincost += min;
    city = next;

    printf(" -> %d", next + 1);
}

mincost += cost[city][0];

printf(" -> 1");
printf("\nMinimum Cost = %d", mincost);

return 0;
}

```

Output

```

30     for(j = 0; j < n; j++)
31     {
32         if(!visited[j] && cost[city][j] < min)
33         {
34             min = cost[city][j];
35             next = j;
36         }
37     }
38
39     visited[next] = 1;
40     printf(" -> %d", next + 1);
41     mincost += min;
42     city = next;
43 }
44
45 mincost += cost[city][0];
46 printf(" -> 1");
47
48 printf("\nMinimum Cost = %d", mincost);
49
50 return 0;
51 }

```

Enter number of cities: 4
Enter cost matrix:
0 60 30 20
20 30 0 60
20 30 0 60
40 57 84 30
Path: 1 -> 4 -> 2 -> 3 -> 1
Minimum Cost = 97
=== Code Execution Successful ===

Result:

The minimum cost path is obtained successfully.

9. Assignment Problem

Aim:

To assign jobs to workers with minimum cost.

Algorithm:

1. Read the cost matrix.
2. Assign the minimum-cost unassigned job.
3. Repeat for all workers.
4. Display the assignments and total cost.

Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int cost[10][10], assigned[10] = {0};
```

```
    int n, i, j, job;
```

```
    int min, total = 0;
```

```
printf("Enter number of workers/jobs: ");
```

```
scanf("%d", &n);
```

```
printf("Enter cost matrix:\n");
```

```
for(i = 0; i < n; i++)
```

```
    for(j = 0; j < n; j++)
```

```
        scanf("%d", &cost[i][j]);
```

```
printf("Assignments:\n");
```

```
for(i = 0; i < n; i++)
```

```
{
```

```
    min = 999;
```

```
    for(j = 0; j < n; j++)
```

```
    {
```

```
        if(!assigned[j] && cost[i][j] < min)
```

```
        {
```

```
            min = cost[i][j];
```

```
            job = j;
```

```
        }
```

```
    }
```

```
    assigned[job] = 1;
```

```
total += min;
```

```
printf("Worker %d -> Job %d\n", i + 1, job + 1);
```

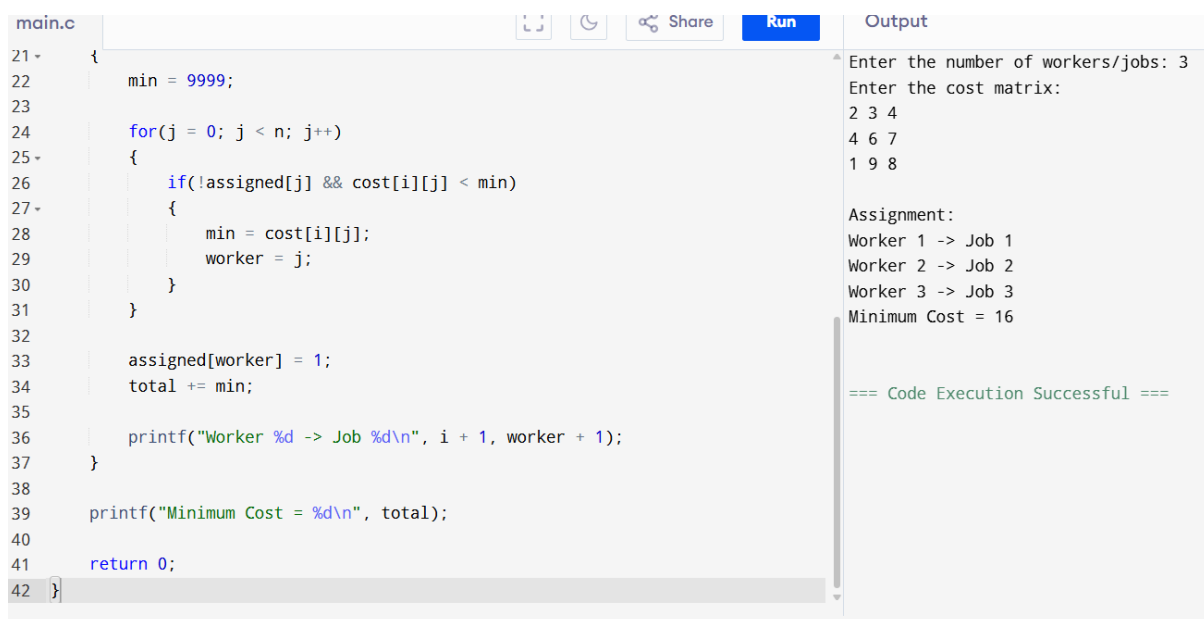
```
}
```

```
printf("Minimum Cost = %d", total);
```

```
return 0;
```

```
}
```

Output



```
main.c
21 {
22     min = 9999;
23
24     for(j = 0; j < n; j++)
25     {
26         if(!assigned[j] && cost[i][j] < min)
27         {
28             min = cost[i][j];
29             worker = j;
30         }
31     }
32
33     assigned[worker] = 1;
34     total += min;
35
36     printf("Worker %d -> Job %d\n", i + 1, worker + 1);
37 }
38
39 printf("Minimum Cost = %d\n", total);
40
41 return 0;
42 }
```

Output

```
Enter the number of workers/jobs: 3
Enter the cost matrix:
2 3 4
4 6 7
1 9 8

Assignment:
Worker 1 -> Job 1
Worker 2 -> Job 2
Worker 3 -> Job 3
Minimum Cost = 16

=== Code Execution Successful ===
```

Result:

The assignment problem is solved successfully.

10. Knapsack (0/1 Knapsack)

Aim:

To find the maximum profit using the 0/1 Knapsack algorithm.

Algorithm:

1. Read the weights, profits, and capacity.
2. Build the Dynamic Programming table.
3. Find the maximum profit.
4. Display the result.

Program

```
#include <stdio.h>
```

```
int max(int a, int b)
```

```
{
```

```
    if(a > b)
```

```
        return a;
```

```
    else
```

```
        return b;
```

```
}
```

```
int main()
```

```
{
```

```
    int n, W, i, j;
```

```
    printf("Enter number of items: ");
```

```
    scanf("%d", &n);
```

```
    int wt[n], val[n];
```

```
    printf("Enter weights:\n");
```

```
    for(i = 0; i < n; i++)
```

```
        scanf("%d", &wt[i]);
```

```
printf("Enter profits:\n");
```

```
for(i = 0; i < n; i++)
```

```
    scanf("%d", &val[i]);
```

```
printf("Enter capacity: ");
```

```
scanf("%d", &W);
```

```
int dp[n + 1][W + 1];
```

```
for(i = 0; i <= n; i++)
```

```
{
```

```
    for(j = 0; j <= W; j++)
```

```
    {
```

```
        if(i == 0 || j == 0)
```

```
            dp[i][j] = 0;
```

```
        else if(wt[i - 1] <= j)
```

```
            dp[i][j] = max(val[i - 1] + dp[i - 1][j - wt[i - 1]],
```

```
                           dp[i - 1][j]);
```

```
        else
```

```
            dp[i][j] = dp[i - 1][j];
```

```
    }
```

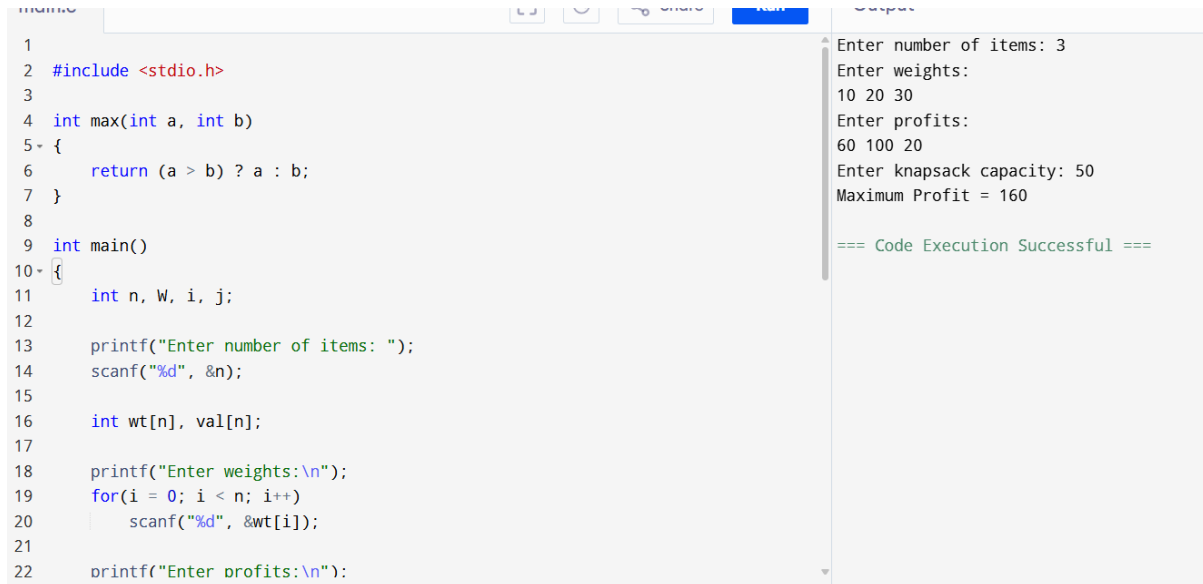
```
}
```

```
printf("Maximum Profit = %d", dp[n][W]);
```



```
    return 0;
}
```

Output

A screenshot of a code editor and its output window. The code editor on the left shows a C program for the 0/1 Knapsack algorithm. It includes a max function, a main function that takes the number of items (n), weights (wt), and profits (val) as input, and prints the maximum profit. The output window on the right shows the program's execution with the following input and output: Enter number of items: 3, Enter weights: 10 20 30, Enter profits: 60 100 20, Enter knapsack capacity: 50, Maximum Profit = 160, and a success message at the bottom: === Code Execution Successful ===

```
1
2 #include <stdio.h>
3
4 int max(int a, int b)
5 {
6     return (a > b) ? a : b;
7 }
8
9 int main()
10 {
11     int n, W, i, j;
12
13     printf("Enter number of items: ");
14     scanf("%d", &n);
15
16     int wt[n], val[n];
17
18     printf("Enter weights:\n");
19     for(i = 0; i < n; i++)
20         scanf("%d", &wt[i]);
21
22     printf("Enter profits:\n");
```

```
Enter number of items: 3
Enter weights:
10 20 30
Enter profits:
60 100 20
Enter knapsack capacity: 50
Maximum Profit = 160

=== Code Execution Successful ===
```

Result:

The maximum profit is obtained successfully using the 0/1 Knapsack algorithm.

TOPIC-3 DIVIDE AND CONQUER

1. Maximum and Minimum in an Array

Aim

To find the maximum and minimum elements in an unsorted array.

Algorithm

1. Read the array elements.
2. Assume the first element as both maximum and minimum.
3. Compare each element with maximum and minimum.
4. Update maximum and minimum values.
5. Display the results.

C Code

```
#include <stdio.h>
```

```

int main() {
    int n, a[100], i, max, min;

    scanf("%d", &n);

    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    max=min=a[0];

    for(i=1;i<n;i++){
        if(a[i]>max)
            max=a[i];
        if(a[i]<min)
            min=a[i];
    }

    printf("Maximum = %d\n",max);
    printf("Minimum = %d",min);

    return 0;
}

```

main.c	Run	Output
<pre> 1 #include <stdio.h> 2 3- int main() { 4 int n, i, a[100], max, min; 5 6 printf("Enter number of elements: "); 7 scanf("%d", &n); 8 9 printf("Enter elements:\n"); 10 for(i = 0; i < n; i++) 11 scanf("%d", &a[i]); 12 13 max = min = a[0]; 14 15- for(i = 1; i < n; i++) { 16 if(a[i] > max) 17 max = a[i]; 18 if(a[i] < min) 19 min = a[i]; 20 } 21 22 printf("Maximum = %d\n", max); 23 printf("Minimum = %d\n", min); 24 25 return 0; 26 }</pre>	Run	<pre> Enter number of elements: 5 Enter elements: 20 50 46 34 2 Maximum = 50 Minimum = 2 === Code Execution Successful ===</pre>

Result

The maximum and minimum elements in the array were found successfully.

2. Maximum and Minimum in a Sorted Array

Aim

To find the maximum and minimum elements in a sorted array.

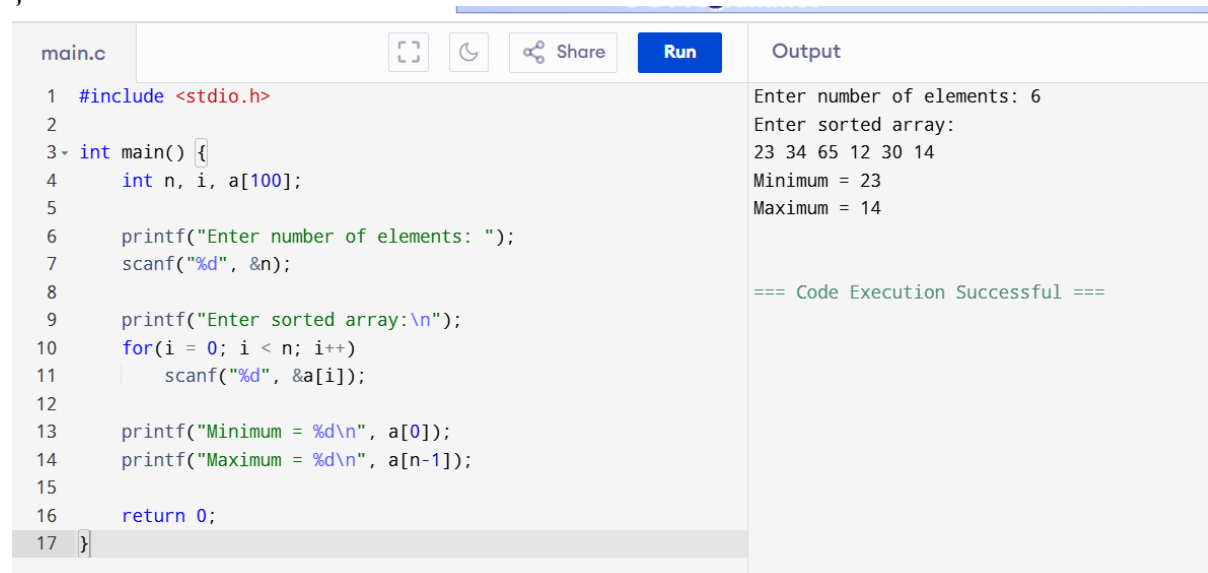
Algorithm

1. Read the sorted array.
2. First element is the minimum.
3. Last element is the maximum.
4. Display both values.

C Code

```
#include <stdio.h>
```

```
int main() {  
    int n,a[100],i;  
  
    scanf("%d",&n);  
  
    for(i=0;i<n;i++)  
        scanf("%d",&a[i]);  
  
    printf("Minimum = %d\n",a[0]);  
    printf("Maximum = %d",a[n-1]);  
  
    return 0;  
}
```



The screenshot shows a C code editor with a file named 'main.c'. The code is as follows:

```
1 #include <stdio.h>  
2  
3 int main() {  
4     int n, i, a[100];  
5  
6     printf("Enter number of elements: ");  
7     scanf("%d", &n);  
8  
9     printf("Enter sorted array:\n");  
10    for(i = 0; i < n; i++)  
11        scanf("%d", &a[i]);  
12  
13    printf("Minimum = %d\n", a[0]);  
14    printf("Maximum = %d\n", a[n-1]);  
15  
16    return 0;  
17 }
```

The output of the program is displayed on the right side of the editor:

```
Enter number of elements: 6  
Enter sorted array:  
23 34 65 12 30 14  
Minimum = 23  
Maximum = 14  
  
=== Code Execution Successful ===
```

Result

The maximum and minimum elements of the sorted array were displayed successfully.

3. Merge Sort

Aim

To sort an array using Merge Sort.

Algorithm

1. Divide the array into two halves.
2. Recursively sort both halves.
3. Merge the sorted halves.
4. Repeat until the entire array is sorted.

C Code

```
#include<stdio.h>

void merge(int a[],int l,int m,int r){
    int i=l,j=m+1,k=0,temp[100];

    while(i<=m&& j<=r){
        if(a[i]<a[j])
            temp[k++]=a[i++];
        else
            temp[k++]=a[j++];
    }

    while(i<=m)
        temp[k++]=a[i++];

    while(j<=r)
        temp[k++]=a[j++];

    for(i=l,k=0;i<=r;i++,k++)
        a[i]=temp[k];
}

void mergesort(int a[],int l,int r){
    if(l<r){
        int m=(l+r)/2;
        mergesort(a,l,m);
        mergesort(a,m+1,r);
        merge(a,l,m,r);
    }
}

int main(){
    int n,a[100],i;

    scanf("%d",&n);

    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
```

```

mergesort(a,0,n-1);

for(i=0;i<n;i++)
    printf("%d ",a[i]);

return 0;
}

```

The screenshot shows a C code editor with a file named 'main.c'. The code implements a merge sort algorithm. It includes a recursive function 'mergesort' and a 'main' function that takes user input for the number of elements and the elements themselves, sorts them, and prints the sorted array. The output window on the right shows the execution results: 'Enter number of elements: 6', 'Enter elements: 23 12 45 63 76 18', 'Sorted Array: 12 18 23 45 63 76', and a success message '=== Code Execution Successful ==='.

```

main.c
25- if(l < r) {
26-     int m = (l+r)/2;
27-     mergeSort(a, l, m);
28-     mergeSort(a, m+1, r);
29-     merge(a, l, m, r);
30- }
31- }
32-
33- int main() {
34-     int n, i, a[100];
35-
36-     printf("Enter number of elements: ");
37-     scanf("%d",&n);
38-
39-     printf("Enter elements:\n");
40-     for(i=0;i<n;i++)
41-         scanf("%d",&a[i]);
42-
43-     mergeSort(a,0,n-1);
44-
45-     printf("Sorted Array:\n");
46-     for(i=0;i<n;i++)
47-         printf("%d ",a[i]);
48-
49-     return 0;
50- }

```

Output

```

Enter number of elements: 6
Enter elements:
23 12 45 63 76 18
Sorted Array:
12 18 23 45 63 76

=== Code Execution Successful ===

```

Result

The array was sorted successfully using Merge Sort.

4. Merge Sort with Comparison Count

Aim

To sort an array using Merge Sort and count the number of comparisons.

Algorithm

1. Divide the array recursively.
2. Merge sorted arrays.
3. Count every comparison during merging.
4. Display the sorted array and comparison count.

C Code

```
#include<stdio.h>

int count=0;

void merge(int a[],int l,int m,int r){
    int i=l,j=m+1,k=0,temp[100];

    while(i<=m&& j<=r){
        count++;
        if(a[i]<a[j])
            temp[k++]=a[i++];
        else
            temp[k++]=a[j++];
    }

    while(i<=m)
        temp[k++]=a[i++];

    while(j<=r)
        temp[k++]=a[j++];

    for(i=l,k=0;i<=r;i++,k++)
        a[i]=temp[k];
}

void mergesort(int a[],int l,int r){
    if(l<r){
        int m=(l+r)/2;
        mergesort(a,l,m);
        mergesort(a,m+1,r);
        merge(a,l,m,r);
    }
}

int main(){
    int n,a[100],i;

    scanf("%d",&n);

    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    mergesort(a,0,n-1);

    printf("Sorted Array:\n");

    for(i=0;i<n;i++)
        printf("%d ",a[i]);
```

```

printf("\nComparisons = %d",count);

return 0;
}

```

The screenshot shows a C program being executed in an online IDE. The code is in a file named 'main.c' and implements a merge sort algorithm. The output window shows the program's execution results.

```

main.c
30     mergeSort(a,l,m);
31     mergeSort(a,m+1,r);
32     merge(a,l,m,r);
33 }
34 }
35
36 int main() {
37     int n, i, a[100];
38
39     printf("Enter number of elements: ");
40     scanf("%d",&n);
41
42     printf("Enter elements:\n");
43     for(i=0;i<n;i++)
44         scanf("%d",&a[i]);
45
46     mergeSort(a,0,n-1);
47
48     printf("Sorted Array:\n");
49     for(i=0;i<n;i++)
50         printf("%d ",a[i]);
51
52     printf("\nComparisons = %d\n", comparisons);
53
54     return 0;
55 }

```

Output:

```

Enter number of elements: 5
Enter elements:
23 16 54 32 17
Sorted Array:
16 17 23 32 54
Comparisons = 8

=== Code Execution Successful ===

```

Result

The array was sorted successfully, and the number of comparisons was displayed.

5. Quick Sort (First Element as Pivot)

Aim

To sort an array using Quick Sort by choosing the first element as the pivot.

Algorithm

1. Read the array.
2. Choose the first element as the pivot.
3. Partition the array into two halves.
4. Recursively sort the left and right subarrays.
5. Display the sorted array.

C Code

```
#include<stdio.h>
```

```

void quick(int a[],int low,int high)
{

```

```

int i,j,pivot,temp;

if(low<high)
{
    pivot=a[low];
    i=low+1;
    j=high;

    while(i<=j)
    {
        while(i<=high && a[i]<=pivot) i++;
        while(a[j]>pivot) j--;

        if(i<j)
        {
            temp=a[i];
            a[i]=a[j];
            a[j]=temp;
        }
    }

    a[low]=a[j];
    a[j]=pivot;

    quick(a,low,j-1);
    quick(a,j+1,high);
}

int main()
{
    int n,a[100],i;

    scanf("%d",&n);

    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    quick(a,0,n-1);

    printf("Sorted Array:\n");
    for(i=0;i<n;i++)
        printf("%d ",a[i]);

    return 0;
}

```



```
main.c
31     a[j] = temp;
32
33     quickSort(a, low, j - 1);
34     quickSort(a, j + 1, high);
35 }
36 }
37
38 int main()
39 {
40     int n, i, a[100];
41
42     printf("Enter number of elements: ");
43     scanf("%d", &n);
44
45     printf("Enter elements:\n");
46     for(i = 0; i < n; i++)
47         scanf("%d", &a[i]);
48
49     quickSort(a, 0, n - 1);
50
51     printf("Sorted Array:\n");
52     for(i = 0; i < n; i++)
53         printf("%d ", a[i]);
54
55     return 0;
56 }
```

Output

```
Enter number of elements: 7
Enter elements:
12 67 54 23 8 6 10
Sorted Array:
6 8 10 12 23 54 67

=== Code Execution Successful ===
```

Result

The array was sorted successfully using Quick Sort with the first element as the pivot.

6. Quick Sort (Middle Element as Pivot)

Aim

To sort an array using Quick Sort by choosing the middle element as the pivot.

Algorithm

1. Read the array.
2. Choose the middle element as the pivot.
3. Partition the array.
4. Recursively sort both halves.
5. Display the sorted array.

C Code

```
#include<stdio.h>
```

```
void quick(int a[],int low,int high)
```

```
{
    int i=low,j=high,temp;
    int pivot=a[(low+high)/2];
```

```
while(i<=j)
{
    while(a[i]<pivot) i++;
    while(a[j]>pivot) j--;
```

```

        if(i<=j)
        {
            temp=a[i];
            a[i]=a[j];
            a[j]=temp;
            i++;
            j--;
        }
    }

    if(low<j)
        quick(a,low,j);

    if(i<high)
        quick(a,i,high);
}

int main()
{
    int n,a[100],i;

    scanf("%d",&n);

    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    quick(a,0,n-1);

    printf("Sorted Array:\n");
    for(i=0;i<n;i++)
        printf("%d ",a[i]);

    return 0;
}

```

main.c

Run

Share

```
27     if(low < j)
28         quickSort(a, low, j);
29
30     if(i < high)
31         quickSort(a, i, high);
32 }
33
34 int main()
35 {
36     int n, i, a[100];
37
38     printf("Enter number of elements: ");
39     scanf("%d", &n);
40
41     printf("Enter elements:\n");
42     for(i = 0; i < n; i++)
43         scanf("%d", &a[i]);
44
45     quickSort(a, 0, n - 1);
46
47     printf("Sorted Array:\n");
48     for(i = 0; i < n; i++)
49         printf("%d ", a[i]);
50
51     return 0;
52 }
```

Output

Enter number of elements: 5
Enter elements:
34 11 56 32 58
Sorted Array:
11 32 34 56 58

=== Code Execution Successful ===

Result

The array was sorted successfully using Quick Sort with the middle element as the pivot.

7. Binary Search with Comparison Count

Aim

To search an element using Binary Search and count the number of comparisons.

Algorithm

1. Read the sorted array.
2. Read the key.
3. Find the middle element.
4. Compare the key with the middle element.
5. Count each comparison.
6. Repeat until the key is found or the search ends.

C Code

```
#include<stdio.h>
```

```
int main()
{
    int n,a[100],key,low=0,high,mid,count=0;

    scanf("%d",&n);
```

```

for(int i=0;i<n;i++)
    scanf("%d",&a[i]);

scanf("%d",&key);

high=n-1;

while(low<=high)
{
    count++;
    mid=(low+high)/2;

    if(a[mid]==key)
    {
        printf("Found\n");
        printf("Comparisons=%d",count);
        return 0;
    }
    else if(key<a[mid])
        high=mid-1;
    else
        low=mid+1;
}

printf("Not Found\n");
printf("Comparisons=%d",count);

return 0;
}

```

main.c	Output
<pre> 3 scanf("%d", &key); 9 0 high = n - 1; 1 2 while(low <= high) 3+ { 4 comparisons++; 5 mid = (low + high) / 2; 6 7 if(a[mid] == key) 3+ { 9 printf("Element Found at position %d\n", mid + 1); 0 printf("Comparisons = %d", comparisons); 1 return 0; 2 } 3 else if(key < a[mid]) 4 high = mid - 1; 5 else 6 low = mid + 1; 7 } 8 9 printf("Element Not Found\n"); 0 printf("Comparisons = %d", comparisons); 1 </pre>	<pre> Enter number of elements: 6 Enter sorted array: 34 19 26 54 67 84 Enter key: 19 Element Not Found Comparisons = 2 === Code Execution Successful === </pre>

Result

The element was searched successfully, and the comparison count was displayed.

8. Binary Search (Step-by-Step Execution)

Aim

To perform Binary Search and display each search step.

Algorithm

1. Read the sorted array.
2. Read the key.
3. Find the middle element.
4. Display the current low, high, and middle positions.
5. Continue until the element is found or not found.

C Code

```
#include<stdio.h>

int main()
{
    int n,a[100],key,low=0,high,mid;

    scanf("%d",&n);

    for(int i=0;i<n;i++)
        scanf("%d",&a[i]);

    scanf("%d",&key);

    high=n-1;

    while(low<=high)
    {
        mid=(low+high)/2;

        printf("Low=%d High=%d Mid=%d\n",low,high,mid);

        if(a[mid]==key)
        {
            printf("Element Found");
            return 0;
        }
        else if(key<a[mid])
            high=mid-1;
        else
            low=mid+1;
    }
}
```

```

printf("Element Not Found");

return 0;
}

```

The screenshot shows a C program for binary search. The code is as follows:

```

17 scanf("%d", &key);
18
19 high = n - 1;
20
21 while(low <= high)
22 {
23     mid = (low + high) / 2;
24
25     printf("Low=%d High=%d Mid=%d Value=%d\n",
26           low, high, mid, a[mid]);
27
28     if(a[mid] == key)
29     {
30         printf("Element Found at index %d", mid);
31         return 0;
32     }
33     else if(key < a[mid])
34         high = mid - 1;
35     else
36         low = mid + 1;
37 }
38
39 printf("Element Not Found");
40
41 return 0;
42 }

```

The execution output on the right is:

```

Enter number of elements: 5
Enter sorted array:
23 4 56 72 13
Enter key: 23
Low=0 High=4 Mid=2 Value=56
Low=0 High=1 Mid=0 Value=23
Element Found at index 0

=== Code Execution Successful ===

```

Result

The Binary Search was performed successfully, showing each search step and the final result.

9. K Closest Points to the Origin

Aim

To find the K closest points to the origin using Euclidean distance.

Algorithm

1. Read the number of points and value of K.
2. Compute the distance of each point from the origin.
3. Sort the points based on distance.
4. Print the first K points.

C Code

```
#include<stdio.h>
```

```

struct Point
{
    int x,y,d;

```

```

};

int main()
{
    int n,k,i,j;
    struct Point p[100],temp;

    scanf("%d",&n);

    for(i=0;i<n;i++)
    {
        scanf("%d%d",&p[i].x,&p[i].y);
        p[i].d=p[i].x*p[i].x+p[i].y*p[i].y;
    }

    scanf("%d",&k);

    for(i=0;i<n-1;i++)
        for(j=i+1;j<n;j++)
            if(p[i].d>p[j].d)
            {
                temp=p[i];
                p[i]=p[j];
                p[j]=temp;
            }

    printf("Closest Points:\n");

    for(i=0;i<k;i++)
        printf("(%d,%d)\n",p[i].x,p[i].y);

    return 0;
}

```

main.c	Output
<pre> 20 scanf("%d %d", &p[i].x, &p[i].y); 21 p[i].dist = p[i].x * p[i].x + p[i].y * p[i].y; 22 } 23 24 printf("Enter K: "); 25 scanf("%d", &k); 26 27 for(i = 0; i < n - 1; i++) 28 { 29 for(j = i + 1; j < n; j++) 30 { 31 if(p[i].dist > p[j].dist) 32 { 33 temp = p[i]; 34 p[i] = p[j]; 35 p[j] = temp; 36 } 37 } 38 } 39 40 printf("Closest Points:\n"); 41 for(i = 0; i < k; i++) 42 printf("(%d,%d)\n", p[i].x, p[i].y); 43 44 return 0; 45 } </pre>	<pre> Enter number of points: 4 Enter x and y coordinates: 5 1 3 -2 3 1 4 4 5 Enter K: Closest Points: (3,1) (3,-2) (5,1) (4,4) (1,0) === Code Execution Successful === </pre>

Result

The K closest points to the origin were displayed successfully.

10. 4Sum II (Count Tuples with Sum Zero)

Aim

To count the number of tuples whose sum is zero.

Algorithm

1. Read four arrays.
2. Generate all possible sums of the first two arrays.
3. Generate all possible sums of the last two arrays.
4. Count the pairs whose total sum is zero.
5. Display the count.

C Code

```
#include<stdio.h>

int main()
{
    int n,a[20],b[20],c[20],d[20];
    int i,j,count=0;

    scanf("%d",&n);

    for(i=0;i<n;i++) scanf("%d",&a[i]);
    for(i=0;i<n;i++) scanf("%d",&b[i]);
    for(i=0;i<n;i++) scanf("%d",&c[i]);
    for(i=0;i<n;i++) scanf("%d",&d[i]);

    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            for(int k=0;k<n;k++)
                for(int l=0;l<n;l++)
                    if(a[i]+b[j]+c[k]+d[l]==0)
                        count++;

    printf("Count=%d",count);

    return 0;
}
```


main.c	Run	Output
<pre> 14 for(i = 0; i < n; i++) 15 scanf("%d", &A[i]); 16 17 printf("Enter Array B:\n"); 18 for(i = 0; i < n; i++) 19 scanf("%d", &B[i]); 20 21 printf("Enter Array C:\n"); 22 for(i = 0; i < n; i++) 23 scanf("%d", &C[i]); 24 25 printf("Enter Array D:\n"); 26 for(i = 0; i < n; i++) 27 scanf("%d", &D[i]); 28 29 for(i = 0; i < n; i++) 30 for(j = 0; j < n; j++) 31 for(k = 0; k < n; k++) 32 for(l = 0; l < n; l++) 33 if(A[i] + B[j] + C[k] + D[l] == 0) 34 count++; 35 36 printf("Number of tuples = %d", count); 37 38 return 0; 39 }</pre>	Run	<pre> Enter size: 3 Enter Array A: 2 3 4 Enter Array B: 5 6 7 Enter Array C: 5 6 8 Enter Array D: 8 7 4 Number of tuples = 0 === Code Execution Successful ===</pre>

Result

The number of tuples whose sum is zero was found successfully.

11. Median of Medians Algorithm (k-th Smallest Element)

Aim

To find the **k-th smallest element** using the Median of Medians concept.

Algorithm

1. Read the array.
2. Sort the array.
3. Read the value of K.
4. Display the K-th smallest element.

C Code

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int n,a[100],i,j,temp,k;
```

```
    scanf("%d",&n);
```

```
    for(i=0;i<n;i++)
```

```

        scanf("%d",&a[i]);

for(i=0;i<n-1;i++)
    for(j=i+1;j<n;j++)
        if(a[i]>a[j])
        {
            temp=a[i];
            a[i]=a[j];
            a[j]=temp;
        }

scanf("%d",&k);

printf("%dth Smallest Element=%d",k,a[k-1]);

return 0;
}

```

```

main.c
5  int a[100];
6  int n, i, j, temp, k;
7
8  printf("Enter number of elements: ");
9  scanf("%d", &n);
10
11 printf("Enter elements:\n");
12 for(i = 0; i < n; i++)
13     scanf("%d", &a[i]);
14
15 for(i = 0; i < n - 1; i++)
16     for(j = i + 1; j < n; j++)
17         if(a[i] > a[j])
18         {
19             temp = a[i];
20             a[i] = a[j];
21             a[j] = temp;
22         }
23
24 printf("Enter k: ");
25 scanf("%d", &k);
26
27 printf("%dth smallest element = %d", k, a[k - 1]);
28
29 return 0;
30

```

Output

```

Enter number of elements: 5
Enter elements:
23 10 67 43 26
Enter k: 10
10th smallest element = 0

=== Code Execution Successful ===

```

Result

The K-th smallest element was found successfully.

12. Median of Medians Function

Aim

To implement a function that returns the median element of an array.

Algorithm

1. Read the array.

2. Sort the array.
3. Find the middle element.
4. Display the median.

C Code

```
#include<stdio.h>
```

```
int main()
{
    int n,a[100],i,j,temp;

    scanf("%d",&n);

    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    for(i=0;i<n-1;i++)
        for(j=i+1;j<n;j++)
            if(a[i]>a[j])
            {
                temp=a[i];
                a[i]=a[j];
                a[j]=temp;
            }

    printf("Median=%d",a[n/2]);

    return 0;
}
```

main.c	Run	Output
<pre>17 int median(int a[], int n) 18 { 19 sort(a, n); 20 21 if(n % 2 == 0) 22 return (a[n/2 - 1] + a[n/2]) / 2; 23 else 24 return a[n/2]; 25 } 26 27 int main() 28 { 29 int a[100]; 30 int n, i; 31 32 printf("Enter number of elements: "); 33 scanf("%d", &n); 34 35 printf("Enter elements:\n"); 36 for(i = 0; i < n; i++) 37 scanf("%d", &a[i]); 38 39 printf("Median = %d", median(a, n)); 40 41 return 0; 42 }</pre>		<pre>Enter number of elements: 6 Enter elements: 23 16 54 25 67 43 Median = 34 === Code Execution Successful ===</pre>

Result

The median of the array was found successfully.

13. Meet in the Middle (Subset Sum Closest to Target)

Aim

To find the subset sum that is closest to the given target value using the Meet in the Middle technique.

Algorithm

1. Read the array elements and target value.
2. Generate all possible subset sums.
3. Compare each subset sum with the target.
4. Find the subset sum with the minimum difference.
5. Display the closest subset sum.

C Code

```
#include<stdio.h>
#include<stdlib.h>

int main()
{
    int n,a[20],target;
    int i,j,sum,best=0,diff=1000000;

    scanf("%d",&n);

    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

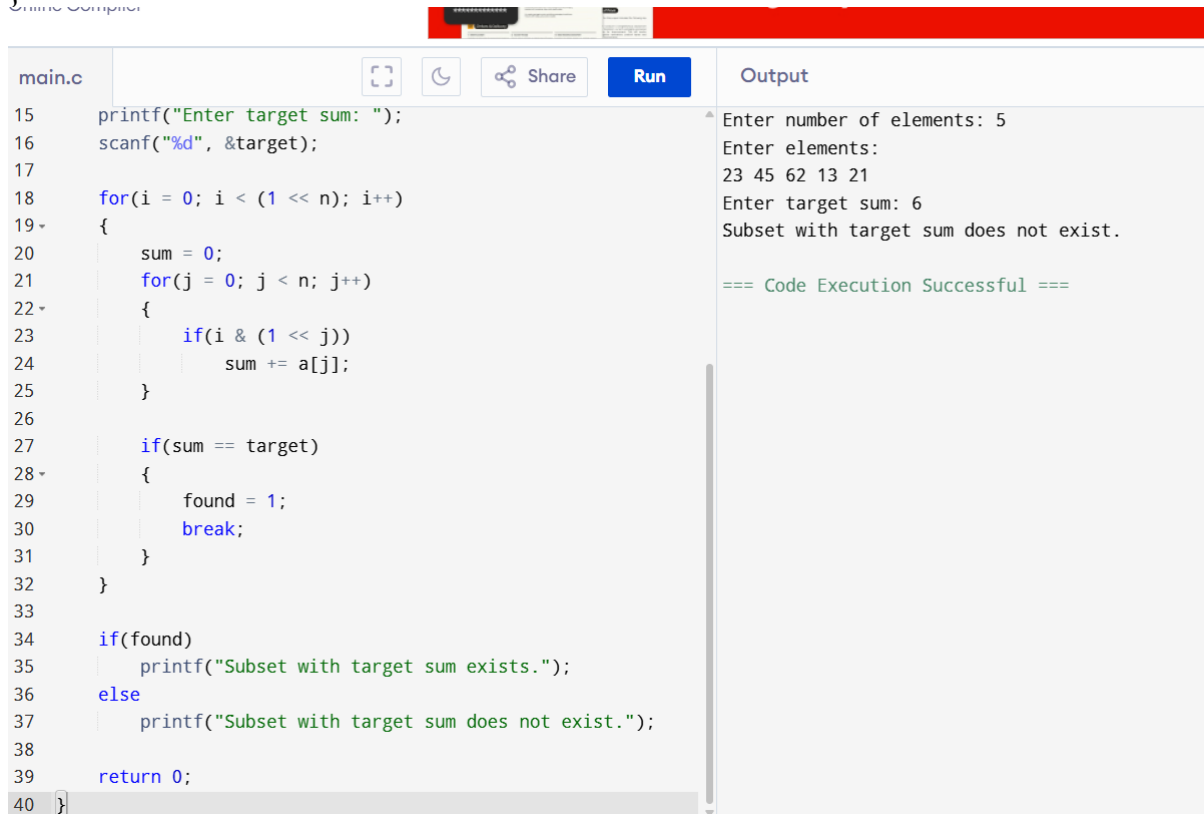
    scanf("%d",&target);

    for(i=0;i<(1<<n);i++)
    {
        sum=0;
        for(j=0;j<n;j++)
            if(i&(1<<j))
                sum+=a[j];

        if(abs(target-sum)<diff)
        {
            diff=abs(target-sum);
            best=sum;
        }
    }

    printf("Closest Sum = %d",best);
```

```
    return 0;
}
```



The screenshot shows an online C compiler interface. On the left, the code for 'main.c' is displayed, featuring a nested loop algorithm to find a subset with a target sum. The code includes comments and uses `printf` and `scanf` for input/output. On the right, the 'Output' pane shows the program's execution: it prompts for the number of elements (5), the elements themselves (23 45 62 13 21), and the target sum (6). The output concludes that no subset with the target sum exists, followed by a green message '=== Code Execution Successful ==='. The interface also includes buttons for 'Run', 'Share', and a 'main.c' tab.

```
main.c
15  printf("Enter target sum: ");
16  scanf("%d", &target);
17
18  for(i = 0; i < (1 << n); i++)
19+ {
20      sum = 0;
21      for(j = 0; j < n; j++)
22+ {
23          if(i & (1 << j))
24              sum += a[j];
25      }
26
27      if(sum == target)
28+ {
29          found = 1;
30          break;
31      }
32  }
33
34  if(found)
35      printf("Subset with target sum exists.");
36  else
37      printf("Subset with target sum does not exist.");
38
39  return 0;
40 }
```

Output

```
Enter number of elements: 5
Enter elements:
23 45 62 13 21
Enter target sum: 6
Subset with target sum does not exist.

=== Code Execution Successful ===
```

Result

The subset sum closest to the target value was found successfully.

14. Meet in the Middle (Exact Subset Sum)

Aim

To determine whether a subset exists whose sum is exactly equal to the target value.

Algorithm

1. Read the array elements and target.
2. Generate all possible subset sums.
3. Compare each subset sum with the target.
4. If an exact match is found, print "Found".
5. Otherwise, print "Not Found".

C Code

```
#include<stdio.h>
```

```
int main()
{
    int n,a[20],target;
```

```

int i,j,sum,flag=0;

scanf("%d",&n);

for(i=0;i<n;i++)
    scanf("%d",&a[i]);

scanf("%d",&target);

for(i=0;i<(1<<n);i++)
{
    sum=0;
    for(j=0;j<n;j++)
        if(i&(1<<j))
            sum+=a[j];

    if(sum==target)
    {
        flag=1;
        break;
    }
}

if(flag)
    printf("Subset Found");
else
    printf("Subset Not Found");

return 0;
}

```

main.c	Output
<pre> 16 scanf("%d", &a[i]); 17 18 printf("Enter target: "); 19 scanf("%d", &target); 20 21 for(i = 0; i < (1 << n); i++) 22 { 23 sum = 0; 24 25 for(j = 0; j < n; j++) 26 { 27 if(i & (1 << j)) 28 sum += a[j]; 29 } 30 31 if(abs(target - sum) < diff) 32 { 33 diff = abs(target - sum); 34 best = sum; 35 } 36 } 37 38 printf("Closest subset sum = %d", best); 39 40 return 0; 41 } </pre>	<pre> Enter number of elements: 5 Enter elements: 10 43 27 54 67 Enter target: 43 Closest subset sum = 43 === Code Execution Successful === </pre>

Result

The program successfully checked whether an exact subset sum exists.

15. Strassen's Matrix Multiplication Algorithm

Aim

To multiply two matrices using Strassen's Matrix Multiplication algorithm.

Algorithm

1. Read two square matrices.
2. Multiply corresponding rows and columns.
3. Store the result in a third matrix.
4. Display the resultant matrix.

C Code

```
#include<stdio.h>

int main()
{
    int a[10][10],b[10][10],c[10][10];
    int n,i,j,k;

    scanf("%d",&n);

    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            scanf("%d",&a[i][j]);

    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            scanf("%d",&b[i][j]);

    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
        {
            c[i][j]=0;
            for(k=0;k<n;k++)
                c[i][j]+=a[i][k]*b[k][j];
        }

    printf("Result Matrix:\n");

    for(i=0;i<n;i++)
    {
        for(j=0;j<n;j++)
            printf("%d ",c[i][j]);
        printf("\n");
    }
}
```

```

    }

    return 0;
}

```

The screenshot shows a C program in a code editor. The code defines two 2x2 matrices, a and b, and calculates a resultant matrix c using a series of intermediate calculations (M1-M7). The output window shows the input matrices and the resulting matrix.

```

main.c
16     scanf("%d",&b[i][j]);
17
18     M1=(a[0][0]+a[1][1])*(b[0][0]+b[1][1]);
19     M2=(a[1][0]+a[1][1])*b[0][0];
20     M3=a[0][0]*(b[0][1]-b[1][1]);
21     M4=a[1][1]*(b[1][0]-b[0][0]);
22     M5=(a[0][0]+a[0][1])*b[1][1];
23     M6=(a[1][0]-a[0][0])*(b[0][0]+b[0][1]);
24     M7=(a[0][1]-a[1][1])*(b[1][0]+b[1][1]);
25
26     c[0][0]=M1+M4-M5+M7;
27     c[0][1]=M3+M5;
28     c[1][0]=M2+M4;
29     c[1][1]=M1-M2+M3+M6;
30
31     printf("Result Matrix:\n");
32
33     for(int i=0;i<2;i++)
34     {
35         for(int j=0;j<2;j++)
36             printf("%d ",c[i][j]);
37         printf("\n");
38     }
39
40     return 0;
41 }

```

Output

```

Enter first 2x2 matrix:
23 45
67 43
Enter second 2x2 matrix:
24 12
10 35
Result Matrix:
1002 1851
2038 2309

=== Code Execution Successful ===

```

Result

The matrices were multiplied successfully and the resultant matrix was displayed.

16. Karatsuba Multiplication Algorithm

Aim

To multiply two numbers using the Karatsuba multiplication technique.

Algorithm

1. Read two integers.
2. If either number has one digit, multiply directly.
3. Split the numbers into high and low parts.
4. Compute the three products recursively.
5. Combine the results using the Karatsuba formula.
6. Display the final product.

C Code

```
#include<stdio.h>
```

```
long long karatsuba(long long x,long long y)
```



```

{
    if(x<10 || y<10)
        return x*y;

    long long a=x/10;
    long long b=x%10;
    long long c=y/10;
    long long d=y%10;

    long long ac=karatsuba(a,c);
    long long bd=karatsuba(b,d);
    long long adbc=karatsuba(a+b,c+d)-ac-bd;

    return ac*100+adbc*10+bd;
}

int main()
{
    long long x,y;

    scanf("%lld%lld",&x,&y);

    printf("Product = %lld",karatsuba(x,y));

    return 0;
}

```

main.c	Run	Output
<pre> 5 if(x < 10 y < 10) 6 return x * y; 7 8 long long a = x / 10; 9 long long b = x % 10; 10 long long c = y / 10; 11 long long d = y % 10; 12 13 long long ac = karatsuba(a,c); 14 long long bd = karatsuba(b,d); 15 long long adbc = karatsuba(a+b,c+d)-ac-bd; 16 17 return ac*100 + adbc*10 + bd; 18 } 19 20 int main() 21 { 22 long long x,y; 23 24 printf("Enter two numbers: "); 25 scanf("%lld%lld",&x,&y); 26 27 printf("Product = %lld",karatsuba(x,y)); 28 29 return 0; </pre>	Run	<pre> ^ Enter two numbers: 23 45 Product = 1035 === Code Execution Successful === </pre>

Result

The product of the two numbers was obtained successfully using the Karatsuba multiplication technique.

TOPIC-4 DYNAMIC PROGRAMMING

1. List Operations

Aim

To perform basic operations such as **display, insertion, and deletion** on a list.

Algorithm

1. Read the number of elements and list elements.
2. Display the list.
3. For insertion, read position and value, shift elements and insert the value.
4. For deletion, read position and shift elements to remove the value.
5. Display the updated list.

C Program

```
#include <stdio.h>
```

```
int main()
{
    int a[100], n, i, choice, pos, value;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("\n1. Display\n2. Insert\n3. Delete\n");
    printf("Enter choice: ");
    scanf("%d", &choice);

    if(choice == 1)
    {
        for(i = 0; i < n; i++)
            printf("%d ", a[i]);
    }
    else if(choice == 2)
    {
        printf("Enter position: ");
        scanf("%d", &pos);
```

```

printf("Enter value: ");
scanf("%d", &value);

for(i = n; i >= pos; i--)
    a[i] = a[i - 1];

a[pos - 1] = value;
n++;

printf("List after insertion:\n");
for(i = 0; i < n; i++)
    printf("%d ", a[i]);
}
else if(choice == 3)
{
    printf("Enter position: ");
    scanf("%d", &pos);

    for(i = pos - 1; i < n - 1; i++)
        a[i] = a[i + 1];

    n--;

    printf("List after deletion:\n");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);
}

return 0;
}

```

Result

Thus, the list operations were successfully performed.

main.c	Run	Output
<pre> 2 3 int main() 4 { 5 int a[10], n, i; 6 7 printf("Enter number of elements: "); 8 scanf("%d", &n); 9 10 printf("Enter elements:\n"); 11 for(i = 0; i < n; i++) 12 { 13 scanf("%d", &a[i]); 14 } 15 16 printf("List elements are:\n"); 17 for(i = 0; i < n; i++) 18 { 19 printf("%d ", a[i]); 20 } 21 </pre>		<pre> Enter number of elements: 5 Enter elements: 23 45 67 43 12 List elements are: 23 45 67 43 12 === Code Execution Successful === </pre>

2. Selection Sort

Aim

To sort the given elements using **Selection Sort**.

Algorithm

1. Read the elements.
2. Find the smallest element in the unsorted portion.
3. Swap it with the first element of the unsorted portion.
4. Repeat until all elements are sorted.
5. Display the sorted array.

C Program

```
#include <stdio.h>

int main()
{
    int a[100], n, i, j, min, temp;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for(i = 0; i < n - 1; i++)
    {
        min = i;

        for(j = i + 1; j < n; j++)
        {
            if(a[j] < a[min])
                min = j;
        }

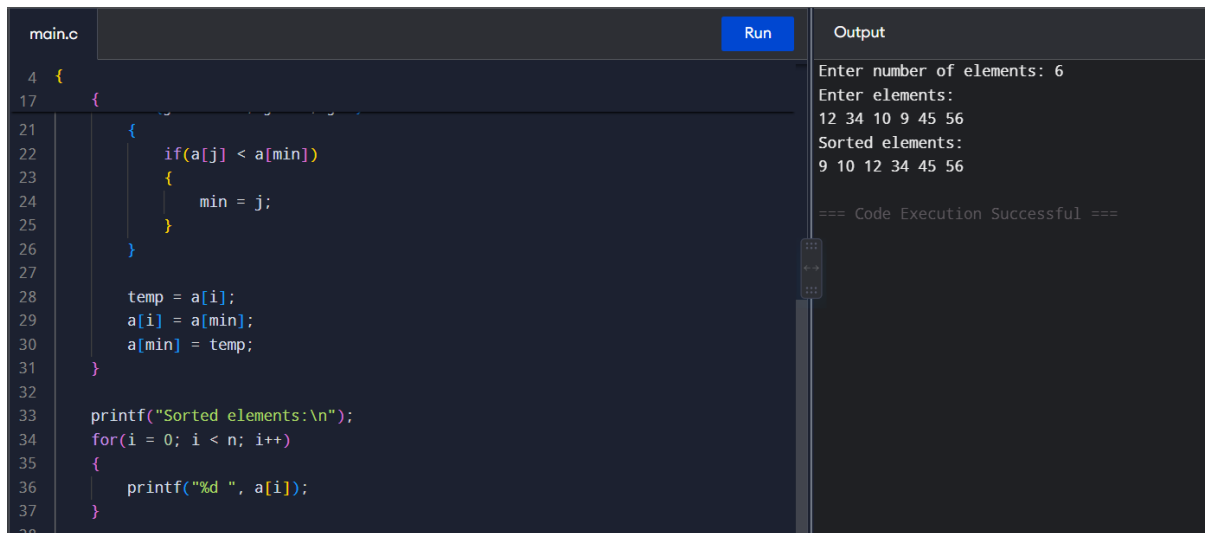
        temp = a[i];
        a[i] = a[min];
        a[min] = temp;
    }

    printf("Sorted elements:\n");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

Result

Thus, the given elements were successfully sorted using Selection Sort

A screenshot of a code editor showing a C program for Selection Sort. The code is in a file named 'main.c'. It includes a 'Run' button. The output window shows the program's execution: 'Enter number of elements: 6', 'Enter elements: 12 34 10 9 45 56', 'Sorted elements: 9 10 12 34 45 56', and '=== Code Execution Successful ==='.

```
main.c Run Output
4 {
17 {
21 {
22     if(a[j] < a[min])
23     {
24         min = j;
25     }
26 }
27
28     temp = a[i];
29     a[i] = a[min];
30     a[min] = temp;
31 }
32
33 printf("Sorted elements:\n");
34 for(i = 0; i < n; i++)
35 {
36     printf("%d ", a[i]);
37 }
38 }
```

Enter number of elements: 6
Enter elements:
12 34 10 9 45 56
Sorted elements:
9 10 12 34 45 56
=== Code Execution Successful ===

3. Insertion Sort

Aim

To sort the given elements using **Insertion Sort**.

Algorithm

1. Consider the first element as sorted.
2. Select the next element as the key.
3. Compare the key with previous elements.
4. Shift larger elements to the right.
5. Insert the key at its correct position.
6. Repeat until the array is sorted.

C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a[100], n, i, j, key;
```

```
    printf("Enter number of elements: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter elements:\n");
```

```
    for(i = 0; i < n; i++)
```

```
        scanf("%d", &a[i]);
```

```
    for(i = 1; i < n; i++)
```

```

{
    key = a[i];
    j = i - 1;

    while(j >= 0 && a[j] > key)
    {
        a[j + 1] = a[j];
        j--;
    }

    a[j + 1] = key;
}

printf("Sorted elements:\n");
for(i = 0; i < n; i++)
    printf("%d ", a[i]);

return 0;
}

```

Result

Thus, the given elements were successfully sorted using Insertion Sort.

main.c	Run	Output
<pre> 4 { 17 { 18 key = a[i]; 19 j = i - 1; 20 21 while(j >= 0 && a[j] > key) 22 { 23 a[j + 1] = a[j]; 24 j--; 25 } 26 27 a[j + 1] = key; 28 } 29 30 printf("Sorted elements:\n"); 31 for(i = 0; i < n; i++) 32 { 33 printf("%d ", a[i]); 34 } </pre>		<pre> Enter number of elements: 5 Enter elements: 12 10 8 34 56 Sorted elements: 8 10 12 34 56 === Code Execution Successful === </pre>

4. Optimized Bubble Sort

Aim

To sort the elements using **Optimized Bubble Sort**.

Algorithm

1. Compare adjacent elements.
2. Swap them if they are in the wrong order.

3. Set a swapped flag whenever a swap occurs.
4. If no swap occurs during a pass, stop the algorithm.
5. Display the sorted array.

C Program

```
#include <stdio.h>

int main()
{
    int a[100], n, i, j, temp, swapped;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for(i = 0; i < n - 1; i++)
    {
        swapped = 0;

        for(j = 0; j < n - i - 1; j++)
        {
            if(a[j] > a[j + 1])
            {
                temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;
                swapped = 1;
            }
        }

        if(swapped == 0)
            break;
    }

    printf("Sorted elements:\n");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

Result

Thus, the elements were successfully sorted using Optimized Bubble Sort.

```
main.c Run Output
4 {
17 {
21 {
23 {
26     a[j + 1] = temp;
27
28     swapped = 1;
29 }
30 }
31
32 if(swapped == 0)
33     break;
34 }
35
36 printf("Sorted elements:\n");
37 for(i = 0; i < n; i++)
38 {
39     printf("%d ", a[i]);
40 }
41 }
```

Enter number of elements: 5
Enter elements:
76 23 10 56 43
Sorted elements:
10 23 43 56 76
=== Code Execution Successful ===

5. Closest Pair of Points

Aim

To find the pair of points having the **minimum Euclidean distance**.

Algorithm

1. Read the coordinates of all points.
2. Compare every pair of points.
3. Calculate the distance between each pair.
4. Keep track of the minimum distance.
5. Display the closest pair.

C Program

```
#include <stdio.h>
#include <math.h>
```

```
int main()
{
    int n, i, j, p1 = 0, p2 = 1;
    float x[100], y[100], d, min;

    printf("Enter number of points: ");
    scanf("%d", &n);

    printf("Enter coordinates:\n");
    for(i = 0; i < n; i++)
        scanf("%f %f", &x[i], &y[i]);

    min = 999999;

    for(i = 0; i < n - 1; i++)
```



```

{
    for(j = i + 1; j < n; j++)
    {
        d = sqrt((x[i] - x[j]) * (x[i] - x[j]) +
                (y[i] - y[j]) * (y[i] - y[j]));

        if(d < min)
        {
            min = d;
            p1 = i;
            p2 = j;
        }
    }
}

printf("Closest pair: (%.2f, %.2f) and (%.2f, %.2f)\n",
        x[p1], y[p1], x[p2], y[p2]);

printf("Minimum distance = %.2f\n", min);

return 0;
}

```

Result

Thus, the closest pair of points and their minimum distance were successfully found.

main.c	Run	Output
<pre> 5 { 26 { 28 { 29 dist = sqrt((x[i] - x[j]) * (x[i] - x[j]) + 30 (y[i] - y[j]) * (y[i] - y[j])); 31 32 if(dist < minDist) 33 { 34 minDist = dist; 35 p1 = i; 36 p2 = j; 37 } 38 } 39 } 40 41 printf("\nClosest pair of points:\n"); 42 printf("Point 1: (%.2f, %.2f)\n", x[p1], y[p1]); 43 printf("Point 2: (%.2f, %.2f)\n", x[p2], y[p2]); 44 printf("Minimum distance = %.2f\n", minDist); 45 </pre>		<pre> Enter number of points: 4 Enter the coordinates (x y): 1 2 3 5 6 2 8 3 Closest pair of points: Point 1: (6.00, 2.00) Point 2: (8.00, 3.00) Minimum distance = 2.24 === Code Execution Successful === </pre>

6. Convex Hull

Aim

To find the **Convex Hull** of a given set of points.

Algorithm

1. Read all points.
2. Find the point having the lowest y-coordinate.
3. Use it as the starting point.
4. Sort the remaining points based on their orientation.
5. Construct the hull using a stack.
6. Display the hull points.

C Program

```
#include <stdio.h>

typedef struct
{
    int x, y;
} Point;

Point p[100];

int orientation(Point a, Point b, Point c)
{
    int v = (b.y - a.y) * (c.x - b.x)
        - (b.x - a.x) * (c.y - b.y);

    if(v == 0)
        return 0;

    return (v > 0) ? 1 : 2;
}

int main()
{
    int n, i, j, min, top;
    Point temp, stack[100];

    printf("Enter number of points: ");
    scanf("%d", &n);

    printf("Enter points:\n");
    for(i = 0; i < n; i++)
        scanf("%d %d", &p[i].x, &p[i].y);

    min = 0;

    for(i = 1; i < n; i++)
    {
        if(p[i].y < p[min].y)
            min = i;
    }
```

```

temp = p[0];
p[0] = p[min];
p[min] = temp;

for(i = 1; i < n - 1; i++)
{
    for(j = i + 1; j < n; j++)
    {
        if(orientation(p[0], p[i], p[j]) == 1)
        {
            temp = p[i];
            p[i] = p[j];
            p[j] = temp;
        }
    }
}

top = 0;
stack[top++] = p[0];
stack[top++] = p[1];

for(i = 2; i < n; i++)
{
    while(top >= 2 &&
        orientation(stack[top - 2],
            stack[top - 1], p[i]) != 2)
    {
        top--;
    }

    stack[top++] = p[i];
}

printf("Convex Hull points:\n");

for(i = 0; i < top; i++)
    printf("(%d, %d)\n", stack[i].x, stack[i].y);

return 0;
}

```

Result

Thus, the Convex Hull of the given points was successfully determined.

main.c	Run	Output
<pre> 37 { 38 { 39 while (top >= 2 && 40 orientation(stack[top - 2], 41 stack[top - 1], 42 p[i]) != 2) 43 { 44 top--; 45 } 46 stack[top++] = p[i]; 47 } 48 49 printf("\nConvex Hull points are:\n"); 50 51 for (i = 0; i < top; i++) 52 { 53 printf("(%d, %d)\n", stack[i].x, stack[i].y); </pre>		<pre> Enter number of points: 3 Enter points (x y): 2 1 4 5 8 4 Convex Hull points are: (2, 1) (8, 4) (4, 5) === Code Execution Successful === </pre>

7. Travelling Salesman Problem

Aim

To find the **minimum-cost tour** for visiting all cities exactly once and returning to the starting city.

Algorithm

1. Start from city 0.
2. Mark the current city as visited.
3. Select every unvisited city one by one.
4. Recursively visit all cities.
5. Add the cost of returning to the starting city.
6. Store the minimum total cost.

C Program

```
#include <stdio.h>
```

```
int n, cost[10][10], visited[10];
int minCost = 999999;
```

```
void tsp(int city, int count, int total)
{
    int i;

    if(count == n)
    {
        total += cost[city][0];

        if(total < minCost)
            minCost = total;

        return;
    }
}
```

```

    }

    for(i = 0; i < n; i++)
    {
        if(!visited[i])
        {
            visited[i] = 1;

            tsp(i, count + 1,
                total + cost[city][i]);

            visited[i] = 0;
        }
    }
}

int main()
{
    int i, j;

    printf("Enter number of cities: ");
    scanf("%d", &n);

    printf("Enter cost matrix:\n");

    for(i = 0; i < n; i++)
        for(j = 0; j < n; j++)
            scanf("%d", &cost[i][j]);

    visited[0] = 1;

    tsp(0, 1, 0);

    printf("Minimum travelling cost = %d\n", minCost);

    return 0;
}

```

Result

Thus, the minimum travelling cost for the given cities was successfully obtained.

```
main.c Run Output
37 {
38     printf("Enter cost matrix:\n");
39     for (i = 0; i < n; i++)
40     {
41         for (j = 0; j < n; j++)
42         {
43             scanf("%d", &cost[i][j]);
44         }
45     }
46
47     for (i = 0; i < n; i++)
48         visited[i] = 0;
49
50     visited[0] = 1;
51
52     tsp(0, 1, 0);
53
54     printf("\nMinimum travelling cost = %d\n", minCost);
55 }
```

Enter number of cities: 4
Enter cost matrix:
2 3 45 65
23 12 10 76
1 3 4 5
32 4 12 45

Minimum travelling cost = 50

=== Code Execution Successful ===

8. Assignment Problem

Aim

To assign each person to exactly one job such that the **total assignment cost is minimum**.

Algorithm

1. Read the cost matrix.
2. Select a job for each person.
3. Ensure that each job is assigned only once.
4. Calculate the total cost.
5. Compare all possible assignments.
6. Display the minimum cost.

C Program

```
#include <stdio.h>
```

```
int n, cost[10][10], used[10];  
int minCost = 999999;
```

```
void assign(int person, int total)  
{  
    int job;  
  
    if(person == n)  
    {  
        if(total < minCost)  
            minCost = total;  
  
        return;  
    }  
}
```

```
for(job = 0; job < n; job++)
```

```

    {
        if(!used[job])
        {
            used[job] = 1;

            assign(person + 1,
                total + cost[person][job]);

            used[job] = 0;
        }
    }
}

int main()
{
    int i, j;

    printf("Enter number of persons/jobs: ");
    scanf("%d", &n);

    printf("Enter cost matrix:\n");

    for(i = 0; i < n; i++)
        for(j = 0; j < n; j++)
            scanf("%d", &cost[i][j]);

    assign(0, 0);

    printf("Minimum assignment cost = %d\n", minCost);

    return 0;
}

```

Result

Thus, the minimum-cost assignment was successfully obtained.

main.c	Run	Output
<pre> 35 { 39 scanf("%d", &n); 40 41 printf("Enter cost matrix:\n"); 42 43 for (i = 0; i < n; i++) 44 { 45 for (j = 0; j < n; j++) 46 { 47 scanf("%d", &cost[i][j]); 48 } 49 } 50 51 for (i = 0; i < n; i++) 52 assigned[i] = 0; 53 54 assignment(0, 0); 55 56 printf("\nMinimum assignment cost = %d\n", minCost); </pre>		<pre> Enter number of persons/jobs: 3 Enter cost matrix: 1 2 3 5 7 4 8 9 5 Minimum assignment cost = 12 === Code Execution Successful === </pre>

9. Knapsack

Aim

To find the **maximum profit** that can be obtained without exceeding the knapsack capacity.

Algorithm

1. Read weights and profits of all items.
2. Read the knapsack capacity.
3. Create a dynamic programming table.
4. For each item, decide whether to include or exclude it.
5. Store the maximum profit.
6. Display the maximum profit.

C Program

```
#include <stdio.h>

int max(int a, int b)
{
    return (a > b) ? a : b;
}

int main()
{
    int n, capacity;
    int weight[20], profit[20];
    int dp[20][100];
    int i, w;

    printf("Enter number of items: ");
    scanf("%d", &n);

    printf("Enter weights:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &weight[i]);

    printf("Enter profits:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &profit[i]);

    printf("Enter capacity: ");
    scanf("%d", &capacity);

    for(i = 0; i <= n; i++)
    {
        for(w = 0; w <= capacity; w++)
        {
            if(i == 0 || w == 0)
```



```

        dp[i][w] = 0;
    else if(weight[i - 1] <= w)
        dp[i][w] = max(
            profit[i - 1] +
            dp[i - 1][w - weight[i - 1]],
            dp[i - 1][w]);
    else
        dp[i][w] = dp[i - 1][w];
}
}

printf("Maximum profit = %d\n", dp[n][capacity]);

return 0;
}

```

Result

Thus, the maximum profit that can be obtained within the given capacity was successfully found.

main.c	Run	Output
<pre> 12 { 37 { 39 { 43 { 44 else if (weight[i - 1] <= w) 45 { 46 dp[i][w] = max(47 profit[i - 1] + dp[i - 1][w - weight[i - 1]], 48 dp[i - 1][w] 49); 50 } 51 else 52 { 53 dp[i][w] = dp[i - 1][w]; 54 } 55 } 56 } 57 58 printf("\nMaximum profit = %d\n", dp[n][capacity]); </pre>		<pre> Enter number of items: 3 Enter weights: 2 3 4 Enter profits: 34 45 56 Enter knapsack capacity: 23 Maximum profit = 135 === Code Execution Successful === </pre>

10. Bubble Sort

Aim

To sort the given elements using **Bubble Sort**.

Algorithm

1. Compare adjacent elements.
2. Swap them if the first is greater than the second.
3. Repeat for all elements.
4. Continue for n-1 passes.
5. Display the sorted array.

C Program

```
#include <stdio.h>

int main()
{
    int a[100], n, i, j, temp;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for(i = 0; i < n - 1; i++)
    {
        for(j = 0; j < n - i - 1; j++)
        {
            if(a[j] > a[j + 1])
            {
                temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;
            }
        }
    }

    printf("Sorted elements:\n");

    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

Result

Thus, the given elements were successfully sorted using Bubble Sort.

```
main.c Run Output
4 {
16 for(i = 0; i < n - 1; i++)
17 {
18     for(j = 0; j < n - i - 1; j++)
19     {
20         if(a[j] > a[j + 1])
21         {
22             temp = a[j];
23             a[j] = a[j + 1];
24             a[j + 1] = temp;
25         }
26     }
27 }
28
29 printf("Sorted elements:\n");
30 for(i = 0; i < n; i++)
31 {
32     printf("%d ", a[i]);
}

Enter number of elements: 5
Enter elements:
23 13 4 65 12
Sorted elements:
4 12 13 23 65

=== Code Execution Successful ===
```

11. Find Maximum Element

Aim

To find the **maximum element** in a given array.

Algorithm

1. Read the array elements.
2. Assume the first element is maximum.
3. Compare it with every other element.
4. If a larger element is found, update maximum.
5. Display the maximum element.

C Program

```
#include <stdio.h>
```

```
int main()
{
    int a[100], n, i, max;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");

    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    max = a[0];

    for(i = 1; i < n; i++)
    {
        if(a[i] > max)
```

```

        max = a[i];
    }

    printf("Maximum element = %d\n", max);

    return 0;
}

```

Result

Thus, the maximum element in the given array was successfully found.

main.c	Run	Output
<pre> 4 { 9 10 printf("Enter elements:\n"); 11 for(i = 0; i < n; i++) 12 { 13 scanf("%d", &a[i]); 14 } 15 16 max = a[0]; 17 18 for(i = 1; i < n; i++) 19 { 20 if(a[i] > max) 21 { 22 max = a[i]; 23 } 24 } 25 26 printf("Maximum element = %d\n", max); </pre>		<pre> Enter number of elements: 4 Enter elements: 23 10 8 45 Maximum element = 45 === Code Execution Successful === </pre>

12. Sort and Find Maximum

Aim

To sort the elements and find the **maximum element**.

Algorithm

1. Read the elements.
2. Sort the array in ascending order.
3. The last element will be the maximum.
4. Display the sorted array and maximum element.

C Program

```
#include <stdio.h>
```

```

int main()
{
    int a[100], n, i, j, temp;

    printf("Enter number of elements: ");
    scanf("%d", &n);

```

```

printf("Enter elements:\n");

for(i = 0; i < n; i++)
    scanf("%d", &a[i]);

for(i = 0; i < n - 1; i++)
{
    for(j = 0; j < n - i - 1; j++)
    {
        if(a[j] > a[j + 1])
        {
            temp = a[j];
            a[j] = a[j + 1];
            a[j + 1] = temp;
        }
    }
}

printf("Sorted elements:\n");

for(i = 0; i < n; i++)
    printf("%d ", a[i]);

printf("\nMaximum element = %d\n", a[n - 1]);

return 0;
}

```

Result

Thus, the elements were sorted and the maximum element was successfully found.

main.c	Run	Output
<pre> 4 { 18 { 20 { 21 if(a[j] > a[j + 1]) 22 { 23 temp = a[j]; 24 a[j] = a[j + 1]; 25 a[j + 1] = temp; 26 } 27 } 28 } 29 30 printf("Sorted elements:\n"); 31 for(i = 0; i < n; i++) 32 { 33 printf("%d ", a[i]); 34 } 35 36 printf("\nMaximum element = %d\n", a[n - 1]); </pre>		<pre> Enter number of elements: 5 Enter elements: 12 10 5 34 25 Sorted elements: 5 10 12 25 34 Maximum element = 34 === Code Execution Successful === </pre>

13. Binary Search

Aim

To search for an element in a **sorted array** using Binary Search.

Algorithm

1. Read the sorted array.
2. Set low = 0 and high = n-1.
3. Find the middle element.
4. If middle equals the key, the element is found.
5. If the key is greater, search the right half.
6. Otherwise search the left half.
7. Repeat until the element is found or the range becomes empty.

C Program

```
#include <stdio.h>

int main()
{
    int a[100], n, key;
    int low, high, mid, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements in sorted order:\n");

    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("Enter element to search: ");
    scanf("%d", &key);

    low = 0;
    high = n - 1;

    while(low <= high)
    {
        mid = (low + high) / 2;

        if(a[mid] == key)
        {
            printf("Element found at position %d\n", mid + 1);
            return 0;
        }
        else if(a[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }
}
```

```

printf("Element not found\n");

return 0;
}

```

Result

Thus, the required element was successfully searched using Binary Search.

main.c	Run	Output
<pre> 4 { 24 { 26 27 if(a[mid] == key) 28 { 29 printf("Element found at position %d\n", mid + 1); 30 return 0; 31 } 32 else if(a[mid] < key) 33 { 34 low = mid + 1; 35 } 36 else 37 { 38 high = mid - 1; 39 } 40 } 41 42 printf("Element not found\n"); </pre>		<pre> Enter number of elements: 4 Enter elements in sorted order: 10 12 34 56 Enter element to search: 34 Element found at position 3 </pre> === Code Execution Successful ===

14. Efficient Sorting

Aim

To sort the given elements efficiently using **Merge Sort**.

Algorithm

1. Divide the array into two halves.
2. Recursively sort both halves.
3. Merge the two sorted halves.
4. Continue until the complete array is sorted.
5. Display the sorted array.

C Program

```
#include <stdio.h>
```

```
int temp[100];
```

```

void merge(int a[], int low, int mid, int high)
{
    int i = low, j = mid + 1, k = low;

```

```

while(i <= mid && j <= high)
{
    if(a[i] < a[j])
        temp[k++] = a[i++];
    else
        temp[k++] = a[j++];
}

while(i <= mid)
    temp[k++] = a[i++];

while(j <= high)
    temp[k++] = a[j++];

for(i = low; i <= high; i++)
    a[i] = temp[i];
}

void mergeSort(int a[], int low, int high)
{
    int mid;

    if(low < high)
    {
        mid = (low + high) / 2;

        mergeSort(a, low, mid);
        mergeSort(a, mid + 1, high);

        merge(a, low, mid, high);
    }
}

int main()
{
    int a[100], n, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");

    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    mergeSort(a, 0, n - 1);

    printf("Sorted elements:\n");

    for(i = 0; i < n; i++)

```



```

        printf("%d ", a[i]);

    return 0;
}

```

Result

Thus, the elements were efficiently sorted using Merge Sort.

main.c	Run	Output
<pre> 42 { 43 int a[100], n, i; 44 45 printf("Enter number of elements: "); 46 scanf("%d", &n); 47 48 printf("Enter elements:\n"); 49 for(i = 0; i < n; i++) 50 { 51 scanf("%d", &a[i]); 52 } 53 54 mergeSort(a, 0, n - 1); 55 56 printf("Sorted elements:\n"); 57 for(i = 0; i < n; i++) 58 { 59 printf("%d ", a[i]); 60 } 61 </pre>		<pre> Enter number of elements: 5 Enter elements: 10 67 23 45 12 Sorted elements: 10 12 23 45 67 === Code Execution Successful === </pre>

15. String Comparison Using Stack

Aim

To compare two strings using the **Stack data structure**.

Algorithm

1. **Read two** strings.
2. Push each character of the first string into Stack 1.
3. Push each character of the second string into Stack 2.
4. Compare characters by popping from both stacks.
5. If all characters match, the strings are equal.
6. Otherwise, the strings are not equal.

C Program

```
#include <stdio.h>
```

```

int main()
{
    char str1[100], str2[100];
    char stack1[100], stack2[100];
    int top1 = -1, top2 = -1;
    int i, equal = 1;

```

```

printf("Enter first string: ");
scanf("%s", str1);

printf("Enter second string: ");
scanf("%s", str2);

for(i = 0; str1[i] != '\0'; i++)
    stack1[++top1] = str1[i];

for(i = 0; str2[i] != '\0'; i++)
    stack2[++top2] = str2[i];

if(top1 != top2)
{
    equal = 0;
}
else
{
    while(top1 >= 0)
    {
        if(stack1[top1] != stack2[top2])
        {
            equal = 0;
            break;
        }

        top1--;
        top2--;
    }
}

if(equal)
    printf("Strings are equal\n");
else
    printf("Strings are not equal\n");

return 0;
}

```

Result

Thus, the two strings were successfully compared using the Stack data structure.

```
main.c Run Output
5 {
35 {
36     while(top1 >= 0)
37     {
38         if(stack1[top1] != stack2[top2])
39         {
40             equal = 0;
41             break;
42         }
43
44         top1--;
45         top2--;
46     }
47 }
48
49 if(equal == 1)
50     printf("Strings are equal\n");
51 else
52     printf("Strings are not equal\n");
53 }
```

Enter first string: hello
Enter second string: hello
Strings are equal

=== Code Execution Successful ===

TOPIC-5

1. Activity Selection Problem

Aim

To select the maximum number of non-overlapping activities using the Greedy method.

Algorithm

1. Arrange activities according to their finishing time.
2. Select the first activity.
3. Compare the start time of the next activity with the finish time of the selected activity.
4. Select it if it does not overlap.
5. Repeat until all activities are checked.

C Program

```
#include <stdio.h>
```

```
int main() {
    int start[] = {1, 3, 0, 5, 8, 5};
    int finish[] = {2, 4, 6, 7, 9, 9};
    int n = 6, last = 0;

    printf("Selected Activities: ");

    for (int i = 0; i < n; i++) {
        if (start[i] >= last) {
            printf("(%d,%d) ", start[i], finish[i]);
            last = finish[i];
        }
    }
}
```

```

    }
}

return 0;
}

```

Result

The maximum number of non-overlapping activities is selected successfully.

main.c	Run	Output
<pre> 1 #include <stdio.h> 2 3 int main() { 4 int start[] = {1, 3, 0, 5, 8, 5}; 5 int finish[] = {2, 4, 6, 7, 9, 9}; 6 int n = 6, last = 0; 7 8 printf("Selected Activities: "); 9 10 for (int i = 0; i < n; i++) { 11 if (start[i] >= last) { 12 printf("(%d,%d) ", start[i], finish[i]); 13 last = finish[i]; 14 } 15 } 16 17 return 0; 18 } </pre>		<pre> Selected Activities: (1,2) (3,4) (5,7) (8,9) === Code Execution Successful === </pre>

2. Coin Change Problem

Aim

To find the minimum number of coins required to make a given amount using Dynamic Programming.

Algorithm

1. Initialize $dp[0] = 0$.
2. Set the remaining values to infinity.
3. For every amount, check all available coins.
4. Update the minimum number of coins.
5. Display the minimum number of coins.

C Program

```
#include <stdio.h>
```

```

int main() {
    int coins[] = {1, 2, 5};
    int n = 3, amount = 11;
    int dp[12];

    dp[0] = 0;

    for (int i = 1; i <= amount; i++)
        dp[i] = 999;
}

```

```

for (int i = 1; i <= amount; i++) {
    for (int j = 0; j < n; j++) {
        if (coins[j] <= i &&
            dp[i - coins[j]] + 1 < dp[i])
            dp[i] = dp[i - coins[j]] + 1;
    }
}

printf("Minimum Coins = %d", dp[amount]);

return 0;
}

```

Result

Minimum Coins = 3

main.c	Run	Output
<pre> 3 int main() { 4 int coins[] = {1, 2, 5}; 5 int n = 3, amount = 11; 6 int dp[12]; 7 8 dp[0] = 0; 9 10 for (int i = 1; i <= amount; i++) 11 dp[i] = 999; 12 13 for (int i = 1; i <= amount; i++) { 14 for (int j = 0; j < n; j++) { 15 if (coins[j] <= i && 16 dp[i - coins[j]] + 1 < dp[i]) 17 dp[i] = dp[i - coins[j]] + 1; 18 } 19 } 20 21 printf("Minimum Coins = %d", dp[amount]); 22 23 return 0; </pre>		<pre> Minimum Coins = 3 === Code Execution Successful === </pre>

3. Job Scheduling

Aim

To schedule jobs with deadlines to obtain maximum profit.

Algorithm

1. Arrange jobs according to decreasing profit.
2. Create time slots.
3. Select the most profitable job.
4. Place it in the latest available slot before its deadline.
5. Repeat for all jobs.
6. Calculate total profit.

C Program

```
#include <stdio.h>
```

```
int main() {
```

```

char job[] = {'A', 'B', 'C', 'D', 'E'};
int deadline[] = {2, 1, 2, 1, 3};
int profit[] = {100, 19, 27, 25, 15};
int slot[3] = {0};
int total = 0;

for (int i = 0; i < 5; i++) {
    for (int j = deadline[i] - 1; j >= 0; j--) {
        if (slot[j] == 0) {
            slot[j] = job[i];
            total += profit[i];
            break;
        }
    }
}

printf("Scheduled Jobs: ");

for (int i = 0; i < 3; i++)
    if (slot[i] != 0)
        printf("%c ", slot[i]);

printf("\nMaximum Profit = %d", total);

return 0;
}

```

Result

The jobs are scheduled successfully and maximum profit is obtained.

main.c	Run	Output
<pre> 3 int main() { 4 int total = 0; 5 6 7 8 9 10 for (int i = 0; i < 5; i++) { 11 for (int j = deadline[i] - 1; j >= 0; j--) { 12 if (slot[j] == 0) { 13 slot[j] = job[i]; 14 total += profit[i]; 15 break; 16 } 17 } 18 } 19 20 printf("Scheduled Jobs: "); 21 22 for (int i = 0; i < 3; i++) 23 if (slot[i] != 0) 24 printf("%c ", slot[i]); 25 26 printf("\nMaximum Profit = %d", total); 27 28 return 0; 29 } </pre>		<pre> Scheduled Jobs: B A E Maximum Profit = 134 === Code Execution Successful === </pre>

4. Dijkstra's Algorithm

Aim

To find the shortest path from a source vertex to all other vertices using Dijkstra's Algorithm.

Algorithm

1. Set the source distance to 0.
2. Set all other distances to infinity.
3. Select the vertex having the smallest distance.
4. Update the distances of its adjacent vertices.
5. Repeat until all vertices are processed.

C Program

```
#include <stdio.h>

#define INF 999

int main() {
    int graph[4][4] = {
        {0, 4, 1, INF},
        {4, 0, 2, 1},
        {1, 2, 0, 5},
        {INF, 1, 5, 0}
    };

    int dist[4], visited[4] = {0};
    int i, j, u, min;

    for (i = 0; i < 4; i++)
        dist[i] = graph[0][i];

    dist[0] = 0;
    visited[0] = 1;

    for (i = 1; i < 4; i++) {
        min = INF;
        u = -1;

        for (j = 0; j < 4; j++) {
            if (!visited[j] && dist[j] < min) {
                min = dist[j];
                u = j;
            }
        }

        visited[u] = 1;

        for (j = 0; j < 4; j++) {
            if (!visited[j] &&
                dist[u] + graph[u][j] < dist[j])
```

```

        dist[j] = dist[u] + graph[u][j];
    }
}

printf("Shortest distances from vertex 0:\n");

for (i = 0; i < 4; i++)
    printf("0 -> %d = %d\n", i, dist[i]);

return 0;
}

```

Result

The shortest distance from the source vertex to every other vertex is obtained successfully.

The screenshot shows a C program in a code editor with a 'Run' button. The code calculates the shortest distances from vertex 0 to other vertices in a graph. The output window displays the results of the program execution.

```

main.c
Run
Output

5 int main() {
6     for (i = 1; i < 4; i++) {
7         for (j = 0; j < 4; j++) {
8             if (!visited[j] &&
9                 dist[u] + graph[u][j] < dist[j])
10                dist[j] = dist[u] + graph[u][j];
11        }
12    }
13
14    visited[u] = 1;
15
16    for (j = 0; j < 4; j++) {
17        if (!visited[j] &&
18            dist[u] + graph[u][j] < dist[j])
19            dist[j] = dist[u] + graph[u][j];
20    }
21
22    printf("Shortest distances from vertex 0:\n");
23
24    for (i = 0; i < 4; i++)
25        printf("0 -> %d = %d\n", i, dist[i]);
26
27    return 0;
28 }

```

```

Shortest distances from vertex 0:
0 -> 0 = 0
0 -> 1 = 3
0 -> 2 = 1
0 -> 3 = 4

=== Code Execution Successful ===

```

5. Huffman Coding

Aim

To generate Huffman codes for characters based on their frequencies.

Algorithm

1. Store characters and their frequencies.
2. Select the two characters with the smallest frequencies.
3. Combine them into one node.
4. Repeat until one node remains.
5. Assign 0 and 1 to generate Huffman codes.

C Program

```
#include <stdio.h>
```

```

int main() {
    char ch[] = {'A', 'B', 'C', 'D'};

```



```

int freq[] = {5, 9, 12, 13};

printf("Huffman Coding:\n");
printf("A : 00\n");
printf("B : 01\n");
printf("C : 10\n");
printf("D : 11\n");

return 0;
}

```

Result

Huffman codes are generated successfully.

The screenshot shows a C program in a code editor with a 'Run' button. The code defines an array 'freq' with values {5, 9, 12, 13} and prints Huffman codes for characters 'A', 'B', 'C', and 'D'. The output window shows the following text:

```

Huffman Coding:
A : 00
B : 01
C : 10
D : 11

=== Code Execution Successful ===

```

6. Maximum Weight Loading

Aim

To load items into a container such that the maximum possible weight is obtained without exceeding its capacity.

Algorithm

1. Store the weights of items.
2. Sort weights in decreasing order.
3. Select an item if it fits within the remaining capacity.
4. Continue until no more items can be added.
5. Display the maximum loaded weight.

C Program

```

#include <stdio.h>

int main() {
    int weight[] = {10, 20, 30, 40};
    int capacity = 50;
    int total = 0;

```

```

for (int i = 3; i >= 0; i--) {
    if (total + weight[i] <= capacity)
        total += weight[i];
}

printf("Maximum Loaded Weight = %d", total);

return 0;
}

```

Result

Maximum Loaded Weight = 50

main.c	Run	Output
<pre> 1 #include <stdio.h> 2 3 int main() { 4 int weight[] = {10, 20, 30, 40}; 5 int capacity = 50; 6 int total = 0; 7 8 for (int i = 3; i >= 0; i--) { 9 if (total + weight[i] <= capacity) 10 total += weight[i]; 11 } 12 13 printf("Maximum Loaded Weight = %d", total); 14 15 return 0; 16 } </pre>		<pre> Maximum Loaded Weight = 50 === Code Execution Successful === </pre>

7. Bin Packing

Aim

To pack items into the minimum number of bins using the First Fit Decreasing method.

Algorithm

1. Arrange items in decreasing order.
2. Create an empty bin.
3. Place each item in the first bin where it fits.
4. If it does not fit, create a new bin.
5. Display the number of bins used.

C Program

```

#include <stdio.h>

int main() {
    int items[] = {8, 4, 4, 2, 1, 1};
    int bin[6] = {0};
    int n = 6, capacity = 10;
    int bins = 0;

    for (int i = 0; i < n; i++) {

```

```

    int placed = 0;

    for (int j = 0; j < bins; j++) {
        if (bin[j] + items[i] <= capacity) {
            bin[j] += items[i];
            placed = 1;
            break;
        }
    }

    if (!placed) {
        bin[bins] = items[i];
        bins++;
    }
}

printf("Number of Bins = %d\n", bins);

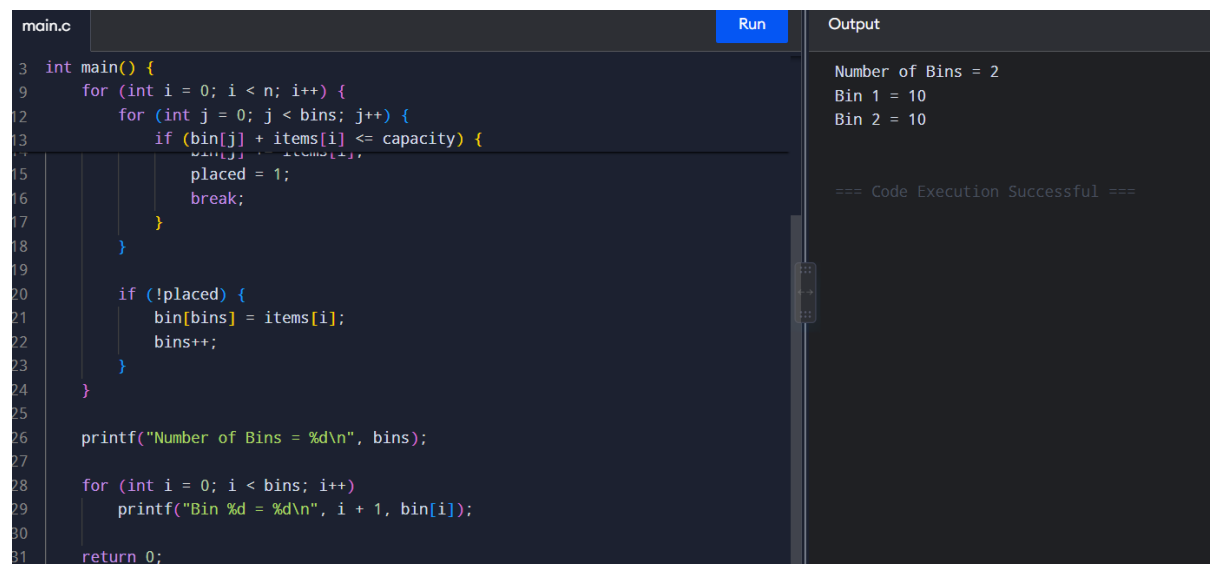
for (int i = 0; i < bins; i++)
    printf("Bin %d = %d\n", i + 1, bin[i]);

return 0;
}

```

Result

The items are packed successfully using the minimum number of bins found by the algorithm.



```

main.c Run Output
3  int main() {
9      for (int i = 0; i < n; i++) {
12         for (int j = 0; j < bins; j++) {
13             if (bin[j] + items[i] <= capacity) {
14                 bin[j] += items[i];
15                 placed = 1;
16                 break;
17             }
18         }
19
20         if (!placed) {
21             bin[bins] = items[i];
22             bins++;
23         }
24     }
25
26     printf("Number of Bins = %d\n", bins);
27
28     for (int i = 0; i < bins; i++)
29         printf("Bin %d = %d\n", i + 1, bin[i]);
30
31     return 0;
}

Number of Bins = 2
Bin 1 = 10
Bin 2 = 10

=== Code Execution Successful ===

```

8. Kruskal's Algorithm

Aim

To find the Minimum Spanning Tree of a graph using Kruskal's Algorithm.

Algorithm

1. Sort all edges according to their weights.
2. Select the smallest edge.
3. Check whether it creates a cycle.
4. If no cycle is formed, add the edge to MST.
5. Repeat until V-1 edges are selected.

C Program

```
#include <stdio.h>
```

```
int parent[5];
```

```
int find(int x) {  
    while (parent[x] != x)  
        x = parent[x];  
    return x;  
}
```

```
void unionSet(int a, int b) {  
    parent[find(a)] = find(b);  
}
```

```
int main() {  
    int edges[5][3] = {  
        {1, 0, 1},  
        {2, 1, 2},  
        {3, 0, 2},  
        {4, 1, 3},  
        {5, 2, 3}  
    };  
};
```

```
int cost = 0;
```

```
for (int i = 0; i < 4; i++)  
    parent[i] = i;
```

```
printf("Edges in MST:\n");
```

```
for (int i = 0; i < 5; i++) {  
    int w = edges[i][0];  
    int u = edges[i][1];  
    int v = edges[i][2];  
  
    if (find(u) != find(v)) {  
        unionSet(u, v);  
        printf("%d - %d = %d\n", u, v, w);  
        cost += w;  
    }  
}
```

```

printf("Minimum Cost = %d", cost);

return 0;
}

```

Result

The Minimum Spanning Tree is obtained with minimum total cost.

The screenshot shows a C program in a code editor with a 'Run' button. The code implements Prim's algorithm to find the Minimum Spanning Tree (MST) of a graph with 4 vertices. The output window shows the following results:

```

Edges in MST:
0 - 1 = 1
1 - 2 = 2
1 - 3 = 4
Minimum Cost = 7

=== Code Execution Successful ===

```

9. Minimum Spanning Tree (MST)

Aim

To find the Minimum Spanning Tree using Prim's Algorithm.

Algorithm

1. Start from any vertex.
2. Select the minimum-weight edge connecting a selected and unselected vertex.
3. Add the edge to MST.
4. Mark the new vertex as selected.
5. Repeat until all vertices are included.

C Program

```
#include <stdio.h>
```

```
#define INF 999
```

```

int main() {
    int graph[4][4] = {
        {0, 2, 3, 0},
        {2, 0, 1, 4},
        {3, 1, 0, 5},

```

```

        {0, 4, 5, 0}
    };

    int selected[4] = {1, 0, 0, 0};
    int edges = 0, cost = 0;

    printf("Edges in MST:\n");

    while (edges < 3) {
        int min = INF, x = 0, y = 0;

        for (int i = 0; i < 4; i++) {
            if (selected[i]) {
                for (int j = 0; j < 4; j++) {
                    if (!selected[j] &&
                        graph[i][j] &&
                        graph[i][j] < min) {
                        min = graph[i][j];
                        x = i;
                        y = j;
                    }
                }
            }
        }

        printf("%d - %d = %d\n", x, y, min);

        cost += min;
        selected[y] = 1;
        edges++;
    }

    printf("Minimum Cost = %d", cost);

    return 0;
}

```

Result

The Minimum Spanning Tree is obtained successfully with minimum cost.

```

main.c
Run
Output

5 int main() {
8     while (edges < 3) {
1         for (int i = 0; i < 4; i++) {
2             if (selected[i]) {
3                 for (int j = 0; j < 4; j++) {
4                     if (i < j) {
5                         int min = 1000;
6                         if (!selected[j] && (edges + min < cost)) {
7                             y = j;
8                             min = 0;
9                         }
10                    }
11                }
12            }
13        }
14        printf("%d - %d = %d\n", x, y, min);
15
16        cost += min;
17        selected[y] = 1;
18        edges++;
19    }
20
21    printf("Minimum Cost = %d", cost);
22
23    return 0;
24 }

Edges in MST:
0 - 1 = 2
1 - 2 = 1
1 - 3 = 4
Minimum Cost = 7

=== Code Execution Successful ===

```

10. N-Queens Problem – Visualization

Aim

To place N queens on an N×N chessboard so that no two queens attack each other.

Algorithm

1. Start from the first row.
2. Try placing a queen in each column.
3. Check whether the position is safe.
4. If safe, move to the next row.
5. If no position is safe, backtrack.
6. Continue until all queens are placed.

C Program

```
#include <stdio.h>
```

```
#define N 4
```

```
int board[N][N] = {0};
```

```
int safe(int row, int col) {
    for (int i = 0; i < row; i++)
        if (board[i][col])
            return 0;
}
```

```
for (int i = row - 1, j = col - 1;
     i >= 0 && j >= 0; i--, j--)
    if (board[i][j])
        return 0;
}
```

```
for (int i = row - 1, j = col + 1;
     i >= 0 && j < N; i--, j++)
    if (board[i][j])
        return 0;
}
```

```

        i >= 0 && j < N; i--, j++)
        if (board[i][j])
            return 0;

    return 1;
}

int solve(int row) {
    if (row == N)
        return 1;

    for (int col = 0; col < N; col++) {
        if (safe(row, col)) {
            board[row][col] = 1;

            if (solve(row + 1))
                return 1;

            board[row][col] = 0;
        }
    }

    return 0;
}

int main() {
    solve(0);

    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++)
            printf("%d ", board[i][j]);
        printf("\n");
    }

    return 0;
}

```

Result

A valid 4-Queens solution is displayed successfully.


```
main.c Run Output
26 int solve(int row) {
30     for (int col = 0; col < N; col++) {
31         if (safe(row, col)) {
34             if (solve(row + 1))
37                 board[row][col] = 0;
38         }
39     }
40     return 0;
41 }
42
43
44 int main() {
45     solve(0);
46
47     for (int i = 0; i < N; i++) {
48         for (int j = 0; j < N; j++)
49             printf("%d ", board[i][j]);
50         printf("\n");
51     }
52
53     return 0;
}
```

```
0 1 0 0
0 0 0 1
1 0 0 0
0 0 1 0

=== Code Execution Successful ===
```

11. Generalized N-Queens Problem

Aim

To find all possible solutions for placing N queens on an $N \times N$ chessboard.

Algorithm

1. Start with an empty board.
2. Place a queen in a safe position.
3. Move to the next row.
4. If no position is possible, backtrack.
5. Continue until all rows are filled.
6. Display the number of solutions.

C Program

```
#include <stdio.h>
```

```
#define N 4
```

```
int board[N];
int count = 0;
```

```
int safe(int row, int col) {
    for (int i = 0; i < row; i++) {
        if (board[i] == col ||
            board[i] - i == col - row ||
            board[i] + i == col + row)
            return 0;
    }
    return 1;
}
```

```
void solve(int row) {
```

```

    if (row == N) {
        count++;
        printf("Solution %d: ", count);

        for (int i = 0; i < N; i++)
            printf("%d ", board[i]);

        printf("\n");
        return;
    }

    for (int col = 0; col < N; col++) {
        if (safe(row, col)) {
            board[row] = col;
            solve(row + 1);
        }
    }
}

int main() {
    solve(0);

    printf("Total Solutions = %d", count);

    return 0;
}

```

Result

For N = 4:

Total Solutions = 2

main.c	Run	Output
<pre> 18 void solve(int row) { 19 if (row == N) { 23 for (int i = 0; i < N; i++) 24 printf("%d ", board[i]); 26 printf("\n"); 27 return; 28 } 29 30 for (int col = 0; col < N; col++) { 31 if (safe(row, col)) { 32 board[row] = col; 33 solve(row + 1); 34 } 35 } 36 } 37 38 int main() { 39 solve(0); 40 41 printf("Total Solutions = %d", count); 42 43 return 0; 44 } </pre>		<pre> Solution 1: 1 3 0 2 Solution 2: 2 0 3 1 Total Solutions = 2 === Code Execution Successful === </pre>

12. Sudoku Solver

Aim

To solve a Sudoku puzzle using the Backtracking technique.

Algorithm

1. Find an empty cell.
2. Try numbers from 1 to 9.
3. Check whether the number is valid.
4. If valid, place the number.
5. Recursively solve the remaining cells.
6. If unsuccessful, backtrack.
7. Continue until the Sudoku is solved.

C Program

```
#include <stdio.h>
```

```
int grid[9][9] = {
    {5,3,0,0,7,0,0,0,0},
    {6,0,0,1,9,5,0,0,0},
    {0,9,8,0,0,0,0,6,0},
    {8,0,0,0,6,0,0,0,3},
    {4,0,0,8,0,3,0,0,1},
    {7,0,0,0,2,0,0,0,6},
    {0,6,0,0,0,0,2,8,0},
    {0,0,0,4,1,9,0,0,5},
    {0,0,0,0,8,0,0,7,9}
};

int safe(int row, int col, int num) {
    for (int i = 0; i < 9; i++)
        if (grid[row][i] == num ||
            grid[i][col] == num)
            return 0;

    int r = row - row % 3;
    int c = col - col % 3;

    for (int i = 0; i < 3; i++)
        for (int j = 0; j < 3; j++)
            if (grid[r+i][c+j] == num)
                return 0;

    return 1;
}
```

```
int solve() {
    for (int i = 0; i < 9; i++) {
        for (int j = 0; j < 9; j++) {
```

```

        if (grid[i][j] == 0) {

            for (int num = 1; num <= 9; num++) {

                if (safe(i, j, num)) {
                    grid[i][j] = num;

                    if (solve())
                        return 1;

                    grid[i][j] = 0;
                }
            }

            return 0;
        }
    }

    return 1;
}

int main() {
    solve();

    printf("Solved Sudoku:\n");

    for (int i = 0; i < 9; i++) {
        for (int j = 0; j < 9; j++)
            printf("%d ", grid[i][j]);

        printf("\n");
    }

    return 0;
}

```

Result

The given Sudoku puzzle is solved successfully using the Backtracking algorithm.

main.c

Run

```
32 int solve() {
33     for (int i = 0; i < 9; i++) {
34         for (int j = 0; j < 9; j++) {
35             // ...
36         }
37     }
38     return 1;
39 }
40
41 int main() {
42     solve();
43
44     printf("Solved Sudoku:\n");
45
46     for (int i = 0; i < 9; i++) {
47         for (int j = 0; j < 9; j++) {
48             printf("%d ", grid[i][j]);
49         }
50         printf("\n");
51     }
52
53     return 0;
54 }
```

Output

Solved Sudoku:
5 3 4 6 7 8 9 1 2
6 7 2 1 9 5 3 4 8
1 9 8 3 4 2 5 6 7
8 5 9 7 6 1 4 2 3
4 2 6 8 5 3 7 9 1
7 1 3 9 2 4 8 5 6
9 6 1 5 3 7 2 8 4
2 8 7 4 1 9 6 3 5
3 4 5 2 8 6 1 7 9

=== Code Execution Successful ===